A Theoretical Analysis of the Velocity Obstacle Method for Nonholonomic Vehicles and Underactuated Surface Vessels

Aurora Haraldsen, *Graduate Student Member, IEEE,* Martin S. Wiig, *Member, IEEE,* Kristin Y. Pettersen, *Fellow, IEEE*

***Abstract*—Collision avoidance systems are pivotal for enabling vehicles to autonomously complete tasks in environments containing obstacles. With its low computational burden and underlying flexibility, the velocity obstacle (VO) algorithm presents a favorable method to avoid collisions, which is based on representing obstacles in the velocity space. In this study, we use the VO principle to form a reactive strategy for vehicles to avoid collisions with dynamic obstacles, which is applied to two different classes of systems, specifically nonholonomic vehicles and underactuated surface vessels. Instead of producing velocity references, the algorithm outputs collision-free directions, thus circumventing the need for explicitly controlling the vehicle speed. Moreover, the majority of existing VO approaches are only supported by simulations and experiments of specific collision avoidance scenarios, and the few studies that include some theoretical assurance are based on assumptions that cannot be guaranteed in the general case. In this paper, we consider factors such as vehicle dynamics and constraints in a rigorous analysis of the algorithm. We analytically derive conditions ensuring feasibility of the avoidance maneuvers and overall safety of the vehicle, which provide intuitive requirements on the parameters of the algorithm. The theoretical results are supported through simulations and experiments of the strategy applied to a nonholonomic vehicle and an underactuated marine vessel.**

I. Introduction

UNCREWED and (semi-)autonomous vehicles are increasingly deployed for operations involving mapping and exploration, transportation, surveillance, intervention, and maintenance. To complete such tasks without human assistance, the vehicle must be capable of avoiding static and moving obstacles along its path. The obstacles can range from other vehicles and physical objects to living things such as animals and people. Safety is thus an immense concern due to the fatal consequences of a collision. Collision avoidance (CA) systems can broadly be split into two main parts: the sensing and tracking of obstacles and the detection and prevention of collisions. This paper exclusively considers the latter.

Recent surveys on CA algorithms can be found in [1]–[3]. Although no precise classification has been established, CA algorithms are often categorized as either global or local approaches, referring to whether the algorithm computes a complete path from one destination to another or just a temporary deviation from the preferred one. Global path planning is typically deployed for finding a collision-free route prior to the mission, considering static information about the environment, while local methods are applied during operation, varying from planning-based methods to purely reactive ones. In this paper, we consider reactive collision avoidance, but the theoretical results are valid also if the strategy is utilized for planning.

Among the existing methods, the Artificial Potential Field (APF) algorithm [4] is a staple of reactive collision avoidance, which is based on a potential field representation of the environment with an attractive part that pulls the vehicle to the goal and a repulsive part that diverts it from obstacles. While the approach is straightforward to implement, it suffers from issues concerning local minima and oscillations as identified in [5], and it only considers static obstacles. [6] presents a method to overcome the appearance of local minima by using a family of 2-dimensional analytical vector fields for navigation in a static obstacle environment, which is proven to almost globally guarantee a safe convergence to the goal.

The study in [7] relates APFs to Control Barrier Functions (CBFs) and demonstrates improved performance when the potential field is formulated under the CBF regime. CBFs present a powerful mathematical tool for enforcing safety in dynamical systems, by ensuring forward invariance of a safe set [8]. Control inputs are chosen by minimizing the deviation between the nominal control and the set of safe inputs, formulated as an optimization problem. A CBF-based approach for avoiding dynamic obstacles was proposed in [9] for fully actuated marine vessels. However, the method builds on the assumption that the obstacle keeps a constant velocity, and underlying actuation constraints are not taken into account.

In view of its many advantages, Model Predictive Control has been suggested for collision avoidance [10], [11], but due to its high computational burden and convergence issues, real-time employment is problematic. A faster alternative is the dynamic window (DW) approach [12] which searches over achievable trajectories in a reduced search space while still incorporating the dynamics of the vehicle. Although the original method only considers first-order nonholonomic constraints and static obstacles, it was extended to second-order models and moving obstacles in modified versions [13], [14]. However, besides increasing the computational cost, there is

This work was supported by the Research Council of Norway through project No. 302435 and the Centres of Excellence funding scheme, project No. 223254 – NTNU AMOS.

Aurora Haraldsen and Kristin Y. Pettersen are with the Department of Engineering Cybernetics, Norwegian University of Science and Technology, N-7491 Trondheim, Norway {Aurora.Haraldsen, Kristin.Y.Pettersen}@ntnu.no

Martin S. Wiig and Kristin Y. Pettersen are with the Norwegian Defence Research Establishment, P.O. Box 25, N-2027 Kjeller, Norway. Martin-Syre.Wiig@ffi.no

also a lack of safety assurance within the modified approaches.

The computationally advantageous velocity obstacle (VO) algorithm [15] is based on representing obstacles in the velocity space; given that the obstacle maintains its current velocity, the VO is defined as the set of vehicle velocities that will eventually lead to a collision between them. The VO representation places no direct requirements on the vehicle system, making it a versatile tool that can be used both reactively and for motion planning. The most noticeable drawbacks in the original algorithm are, however, the assumptions that the velocities are constant over discrete time intervals and that the avoidance maneuvers are given as linear velocity changes which may be infeasible with respect to the vehicle kinematics.

An extensive amount of research has been published on the use of the VO concept for collision avoidance in 2D. The approach [16] presents a generalization of the VO concept to incorporate the constraints of a car-like robot in order to compute feasible control actions in a multi-obstacle environment. However, safety is established under the assumption that the obstacles move as predicted. The Acceleration VO (AVO) [17] modifies the algorithm to include acceleration constraints and introduces proportional control of the acceleration to make smooth transitions between consecutive velocities. The AVO guarantees avoidance if the obstacle keeps a constant velocity for a chosen time horizon. The Reciprocal VO (RVO) [18] presents a distributed method for multi-agent collision avoidance, by imposing that all agents act according to the VO of other agents in a close neighborhood. The AVO and RVO are combined into Optimal Reciprocal Collision Avoidance (ORCA) in [17]. The VO concept was extended to 3D in [19], and it has been applied to systems such as surface and underwater vehicles [20]–[24] and aerial vehicles [25]–[27].

This paper is motivated by the lack of a theoretical foundation of the VO method when applied to dynamical systems in general obstacle encounters. Although a great deal of variations of the method have been proposed, a vast majority are validated only in simulations or experiments that do not provide any common guarantees of safety. In the few studies that provide some theoretical perspectives on the collision avoidance property, the analyses rely on assumptions that are not realistic in a general setting, e.g. that obstacles move with piece-wise constant velocities, and they either fully or partly neglect the vehicle dynamics and constraints. In this paper, we present theoretical analyses of the collision avoidance problem for the class of nonholonomic, kinematic vehicles and underactuated marine vessels. Through these studies, we provide analytical proofs and explicit conditions for safety, which are then demonstrated both through simulations and experiments. The studies specifically consider the implications of the vehicle kinematics and dynamics on the collision avoidance trajectories, and an obstacle is modeled as a kinematic vehicle with potentially varying speed and heading to provide rigorous guarantees for safety.

The proposed algorithm is based on simple and intuitive avoidance principles and requires limited obstacle information, motivated by the work on the related Constant Avoidance Angle (CAA) algorithm [28]–[30]. While the underlying principle of the CAA is geometrically similar to the VO, the algorithm necessitates that the vehicle keeps a constant forward speed, and it can only guarantee safety for vehicles that are capable of instantly changing their turning direction when the obstacle is in close range. The proposed approach both addresses and alleviates these limitations. Based on the VO principle, we implement a sense-and-avoid approach, maintaining the vehicle's intended (nominal) course until an impending collision is detected. When an evasive maneuver is triggered, the vehicle is steered in a safe direction until the system can securely resume its nominal behavior. The nominally desired control objectives are in this study formulated as target reaching and path following, as these are typically required behaviors of autonomous vehicles. There are, however, no requirements on the nominal goal, which makes the proposed method suitable for many applications. Furthermore, to avoid collision, the vehicle is only required to keep a certain direction, and thus the speed of the vehicle can be controlled separately under some mild assumptions on the acceleration and minimum vehicle speed. Due to this flexibility, the strategy can combined with most existing control regimes.

Preliminary results have been published in [31]–[33], where the velocity obstacle and collision cone principles have been studied for nonholonomic, unicycle-type vehicles and underactuated surface vessels. In this work, we further generalize and improve these results, obtaining more accurate criteria for safety and thus less conservative bounds on the parameters of the algorithm. In [31], the avoidance strategy was introduced and formal safety guarantees were derived for nonholonomic vehicles with constant forward speeds. This analysis was extended to include a nonzero forward acceleration in [33], where we also experimentally verified the method. In this paper, we bridge the findings from the study of nonholonomic vehicles to the more intricate category of underactuated marine vehicles. Furthermore, we showcase the algorithm through simulations in more demanding scenarios involving multiple moving obstacles. For underactuated surface vessels, this paper presents a significant enhancement of both the theory and validation provided in [32]. Whereas the preliminary study restricted the vehicle to keep a constant surge speed, we here consider the more general scenario where the vehicle keeps a potentially nonzero surge acceleration. Moreover, the safety conditions found in [32] are shown to be overly conservative and more precise bounds are obtained, which can then be directly related to the conditions found for unicycles. While the preliminary work only includes numerical simulations, the theoretical results are here additionally supported by experiments with an underactuated marine vessel.

The paper is organized as follows. Section II defines the velocity obstacle for two moving objects. This principle is leveraged in Section III, where we outline the proposed collision avoidance strategy. The algorithm is studied for nonholonomic vehicles in Section IV. We define the control objectives and propose the control system for the achievement of these goals. Moreover, the complete system is analyzed, taking into consideration both the nonholonomic constraints and additional input limits. Multi-obstacle simulations are presented. In Section V, we extend the avoidance strategy

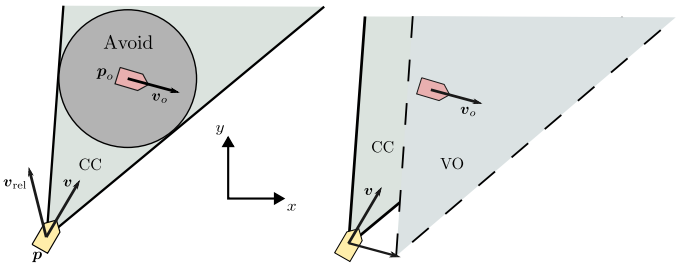


Fig. 1. A potential collision between the vehicle and an obstacle. The collision cone (CC) in (a) comprises the velocities of the vehicle relative to the obstacle that guarantee a collision, assuming the relative velocity is constant. The velocity obstacle (VO) in (b) encapsulates the corresponding absolute vehicle velocities, where the CC has been adjusted for the obstacle's velocity. In the above scenario, the current velocities do not result in a future collision.

to underactuated surface vessels, where we present the full control design and analyze the method in closed-loop with the underactuated dynamics. The theoretical results are supported by simulations and full-scale experiments with a marine vessel. The concluding remarks are given in the last section.

II. The Velocity Obstacle

This section briefly introduces the velocity obstacle principle that facilitates the design of the collision avoidance algorithm presented in the next section. We begin by introducing some notation: let $\boldsymbol{p}$ and $\boldsymbol{v} \triangleq \dot{\boldsymbol{p}}$ denote linear position and velocity, respectively, and define the 2-norm as $\|(\cdot)\|$. For 2D vectors, the operator $\angle$ represents the transformation $\text{atan2}(\cdot,\cdot)$. We examine a vehicle and an obstacle moving in the horizontal plane, denoting the obstacle with the subscript o. Consider the configuration in Figure 1, displaying the vehicle in yellow and the obstacle in red. Define the distance between the vehicle and the obstacle as $d_o \triangleq \|\boldsymbol{p} - \boldsymbol{p}_o\|$.

Definition II.1. *A collision occurs when*

$$d_o < d_{\min}, \tag{1}$$

where $d_{\min} > 0$ is the minimum required distance between the vehicle and the obstacle.

In practice, the minimum distance, $d_{\min}$, serves as a design parameter that compensates for the physical space occupied by the vehicle and the obstacle. The advantage of this formulation is that we can treat the vehicle as a point and the obstacle as the moving region

$$D_{\text{col}} \triangleq \left\{\boldsymbol{p} \in \mathbb{R}^2 : d_o < d_{\min}\right\}, \tag{2}$$

shown as the shaded disk in Figure 1a. The *collision cone*, abbreviated as CC, is defined as the set of relative velocities between the vehicle and the obstacle, $\boldsymbol{v}_{\text{rel}} \triangleq \boldsymbol{v} - \boldsymbol{v}_o$, that will result in a collision (i.e., that the vehicle ends up inside of the collision region D_{col}) at some future time. Note that a collision is only *guaranteed* if $\boldsymbol{v}_{\text{rel}}$ is constant; an illustration of the CC set can be seen in Figure 1. The *velocity obstacle* (VO) is the corresponding set of all vehicle velocities, $\boldsymbol{v}$, that will result in a collision with the obstacle moving at constant velocity $\boldsymbol{v}_o$. The VO is obtained by adding the obstacle velocity $\boldsymbol{v}_o$ to each vector in the CC. Geometrically, this operation is equivalent to shifting the CC by the vector $\boldsymbol{v}_o$, as seen in Figure 1b. If the vehicle and the obstacle remain at their current velocities, a collision will thereby occur in the future if the vehicle's absolute velocity points within the VO set. While in this work, we do not require the velocities to be constant, the condition is still an indication that the vehicle and an obstacle are headed for collision. This situation will be referred to as a *conflict*.

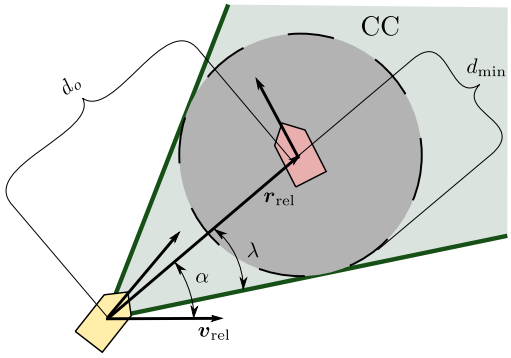


Fig. 2. Geometry of the collision cone. The half-angle of the cone is given by λ, and α represents the angle between the relative velocity vector, $\boldsymbol{v}_{\text{rel}}$, and the line $\boldsymbol{r}_{\text{rel}}$ going from the vehicle center to the obstacle center. A conflict occurs if the angle α becomes smaller than the collision cone's half-angle.

Having established the main concepts, we will next adapt them to the more general scenario where the velocities may be time-varying. Consider Figure 2; the half-angle of the collision cone can be found geometrically as

$$\lambda \triangleq \arcsin\left(\frac{d_{\min}}{d_o}\right), \tag{3}$$

and the angle between the relative velocity vector, $\boldsymbol{v}_{\text{rel}}$, and the relative position vector, $\boldsymbol{r}_{\text{rel}} \triangleq \boldsymbol{p}_o - \boldsymbol{p}$, is found as

$$\alpha \triangleq \arccos\left(\frac{\boldsymbol{v}_{\text{rel}}^\top \boldsymbol{r}_{\text{rel}}}{\|\boldsymbol{v}_{\text{rel}}\|\|\boldsymbol{r}_{\text{rel}}\|}\right). \tag{4}$$

The following condition implies that the relative velocity lies within the collision cone:

$$\alpha < \lambda. \tag{5}$$

While condition (5) provides a convenient metric to determine the presence of a conflict, it can similarly be used to establish the absence of future collisions:

Lemma II.1. *Given that the vehicle is not in a collision at time $t_0 \geq 0$, if*

$$\alpha(t) \geq \lambda(t), \quad \forall t \geq t_0, \tag{6}$$

then the vehicle will avoid a collision,

$$d_o(t) \geq d_{\min}, \quad \forall t \geq t_0. \tag{7}$$

Proof. The time-evolution of the distance, d_o, is found geometrically as

$$\dot{d}_o = -\|\boldsymbol{v}_{\text{rel}}\| \cos(\alpha). \tag{8}$$

Applying the inequality (6) and the relations (3) – (4), (8) satisfies

$$\dot{d}_o \geq -\frac{\|\boldsymbol{v}_{\text{rel}}\|}{d_o}\sqrt{d_o^2 - d_{\min}^2}. \tag{9}$$

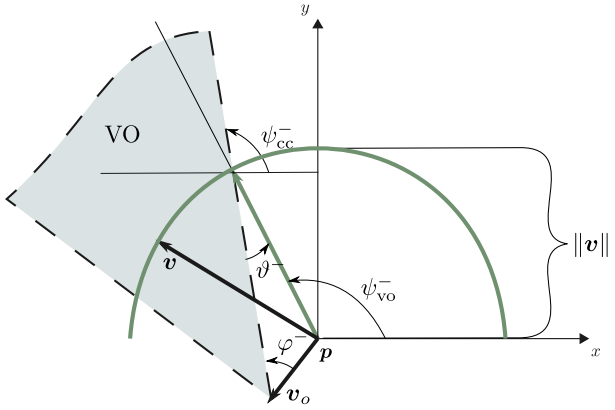


Fig. 3. Illustration of the angles φ^- and ϑ^-. The green half-circle represents the vehicle velocity at different directions when moving at the current speed. The velocity vector touches the right edge of the VO at an angle of ψ_{vo}^{-}, as marked in green.

Since $d_o(t_0) \geq d_{\min}$ and $d_o = d_{\min} \implies \dot{d}_o(t) \geq 0$ for all $t \geq t_0$ by (9), it follows that the distance to the obstacle remains larger than or equal to $d_{\min}$. □

In case of a conflict, the VO provides a more practical means of determining the necessary maneuver compared to the collision cone. The primary focus lies in identifying at which vehicle directions, $\angle \boldsymbol{v}$, the velocity vector remains on the outside of the VO. Let $\psi_{\mathrm{cc}}^{\pm} \triangleq \angle \boldsymbol{r}_{\mathrm{rel}} \pm \lambda$ be the directions of the left $(+)$ and the right $(-)$ edge of the collision cone. At the given speed $\|\boldsymbol{v}\|$, the vehicle's velocity vector aligns with the corresponding edge of the VO when the direction of this vector equals

$$\psi_{\mathrm{vo}}^{\pm} \triangleq \psi_{\mathrm{cc}}^{\pm} + \vartheta^{\pm}, \tag{10}$$

as illustrated in Figure 3, where

$$\vartheta^{\pm} \triangleq \arcsin\left(\frac{\|\boldsymbol{v}_o\|}{\|\boldsymbol{v}\|}\sin(\varphi^{\pm})\right), \quad \varphi^{\pm} \triangleq \pi - \angle \boldsymbol{v}_o + \psi_{\mathrm{cc}}^{\pm}. \tag{11}$$

Unsafe vehicle directions (i.e., those constituting a conflict) lie in the interval between the right and the left angle, whereas all directions outside of this interval are non-conflicting.

Note that the angles (10) are found at the intersections between the circle centered at $\boldsymbol{p}$ with radius equal to the magnitude of $\boldsymbol{v}$ and the edges of the VO. In fact, this circle does not always intersect these lines, particularly when the vehicle speed is too low compared to the obstacle's speed, such that $\vartheta^{\pm}$ are undefined. A practical consequence of this situation is that the vehicle may be unable to resolve an ongoing conflict by changing its direction. To prevent the vehicle from reaching such configurations, we introduce minimum requirements on the vehicle's speed in Sections IV and V.

III. The Velocity Obstacle Collision Avoidance Strategy

In this section, we apply the principles established in the previous section to develop a strategy for avoiding collisions with obstacles whilst executing a nominal task. The algorithm we propose operates at the guidance level, where it takes a nominal desired direction, ψ_{nom}, as input and produces a potentially adjusted and safe reference direction, ψ_{d}. We first introduce the model of a dynamic obstacle that will serve as the basis for our study. Then we outline the conditions that dictate when the vehicle should execute an evasive maneuver in the presence of an obstacle, deviating from the nominal direction. Lastly, we define the avoidance direction to be followed during the execution of the maneuver, and we present the condition that concludes the maneuver, allowing the vehicle to return to its nominal trajectory.

A. Obstacle Model

We will consider an obstacle of radius R_o described by the kinematic equations:

$$\dot{\boldsymbol{p}}_o = u_o\left[\cos\psi_o,\, \sin\psi_o\right]^\top, \tag{12a}$$

$$\dot{\psi}_o = r_o, \qquad r_o \in [-r_{o\max},\, r_{o\max}], \tag{12b}$$

$$\dot{u}_o = a_o, \qquad a_o \in [-a_{o\max},\, a_{o\max}], \tag{12c}$$

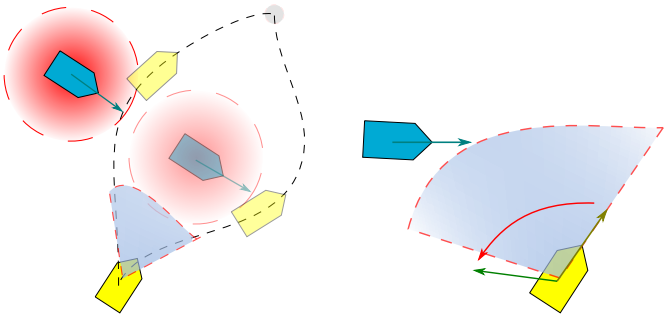
where ψ_o is the orientation (heading), $u_o \in [0, u_{o\max}]$ is the forward speed, r_o is the turning rate, and a_o is the forward acceleration. The parameters $r_{o\max} \geq 0$, $a_{o\max} \geq 0$, and $u_{o\max} \geq 0$ represent limits to the obstacle dynamics. This model captures the generic motion characteristics of a moving object as well as the kinematics of a wide range of vehicles. We assume that the sensing system of the vehicle accurately measures the position $\boldsymbol{p}_o \triangleq [x_o,\, y_o]^\top$ and linear velocity $\boldsymbol{v}_o \triangleq \dot{\boldsymbol{p}}_o$ of the obstacle.

B. Conflict Detection

A conflict indicates that there will be a future collision with the obstacle if the current velocity is kept, thus providing an intuitive condition for initiating an evasive maneuver. Noting, however, that the conflict has to persist up until the time of collision, it becomes conservative to react to potential collisions that are still far from occurring. As such, it is common practice to prioritize avoiding collision based on imminence. To determine whether a collision is sufficiently close to act on, we define a safety distance to the obstacle, $d_{\mathrm{safe}} > 0$, such that beyond this distance, conflicts are neglected.

If the distance to the obstacle decreases below this safety threshold, we evaluate the necessity of an evasive maneuver based on the nearness to a conflict, rather than a conflict itself. This adds an extra layer of safety to account for the possibility that the evasive maneuver may require a complete shift in turning direction. Moreover, this design choice will be of importance for guaranteeing safety when the strategy is applied to vehicles with significant dynamics, like marine surface vessels. To this end, we increase the angular width of the collision cone by adding an angle $\psi_{\mathrm{safe}} \in [0, \frac{\pi}{2})$ to each side, yielding $\lambda_{\mathrm{safe}} \triangleq \lambda + \psi_{\mathrm{safe}}$. Since we are interested in whether the nominal direction leads to collision, we compare λ_{safe} to the angle between the nominal relative velocity, $\boldsymbol{v}_{\mathrm{nom,rel}} \triangleq \boldsymbol{v}_{\mathrm{nom}} - \boldsymbol{v}_o$, and the line $\boldsymbol{r}_{\mathrm{rel}}$, where $\boldsymbol{v}_{\mathrm{nom}}$ is obtained from ψ_{nom} through the appropriate mapping. Defining $\alpha_{\mathrm{nom}} \triangleq \arccos\left(\frac{\boldsymbol{v}_{\mathrm{rel,nom}}^\top \boldsymbol{r}_{\mathrm{rel}}}{\|\boldsymbol{v}_{\mathrm{rel,nom}}\|\|\boldsymbol{r}_{\mathrm{rel}}\|}\right)$, the required conditions for initiating an avoidance maneuver can thus be summarized as

$$\alpha_{\mathrm{nom}} < \lambda_{\mathrm{safe}}, \quad d_o \leq d_{\mathrm{safe}}. \tag{13}$$



(a) (b)

Fig. 4. (a) Given that the obstacle remains at its current velocity, a maneuver can be characterized according to the resulting relative movement of the vehicle with respect to the obstacle. In this case, the choices are to go behind or in front of the obstacle, corresponding to a velocity along the left or the right edge of the VO, respectively. The former option is generally the most intuitive and potentially the safest. Above, the shaded, blue sector represents directions between ψ_{vo}^{-} and ψ_{vo}^{+}. (b) A situation in which the nominal heading (green) is safe, but the shortest turn (red) could lead to a collision.

C. Avoidance Maneuver

The velocity obstacle strategy offers a set of potential avoidance maneuvers, providing a significant amount of flexibility in shaping the resulting trajectory. Noting that the edges (10) define the minimum required angles to avoid the obstacle, we select the avoidance direction along one of these edges parameterized by an angle $\delta > 0$. By increasing δ, the vehicle keeps a proportionally larger distance to the obstacle while passing. Hence, once (13) is triggered, we let $\psi_{\text{d}} = \psi_{\text{ca}}^{j}$ with

$$\psi_{\text{ca}}^{\pm} \triangleq \psi_{\text{vo}}^{\pm} \pm \delta, \tag{14}$$

where the parameter $j \in \{\pm\}$ provides another degree of flexibility. Figure 4a illustrates how the choice of j impacts the resulting trajectory of the vehicle relative to the obstacle. To further lower the collision risk, passing behind an obstacle can prove beneficial [30]. Therefore, if the avoidance maneuver is initiated at a safe distance from the obstacle such that $d_o = d_{\text{safe}}$, we choose the direction farthest from the direction that the obstacle is moving in:

$$\underset{j\in\{\pm\}}{\operatorname{argmax}} \left|\psi_{\text{ca}}^{j} - \angle \boldsymbol{v}_o\right|, \tag{15}$$

where the angular difference is mapped to the interval $(-\pi, \pi]$. Otherwise, we choose the closest avoidance direction:

$$\underset{j\in\{\pm\}}{\operatorname{argmin}} \left|\psi_{\text{ca}}^{j} - \angle \boldsymbol{v}\right|, \tag{16}$$

which is to prevent the vehicle from entering into a conflict with the obstacle when they are in close proximity.

Once the nominally desired direction is farther from a conflict than the angular safe distance, i.e.:

$$\alpha_{\text{nom}} \geq \lambda_{\text{safe}}, \tag{17}$$

the avoidance maneuver is terminated and the nominal behaviour is resumed by letting $\psi_{\text{d}} = \psi_{\text{nom}}$. Note, however, that it is possible that the nominal velocity is directed along the opposite edge of the VO compared to that of the kept velocity, in which case turning towards the nominal direction could make the vehicle collide with the obstacle (cf. Figure 4b). Safety can in this case be ensured by requiring the vehicle to take the longest turn. We note that such an implementation will be implicitly assumed throughout this study.

Remark 1. *If there are multiple obstacles, the independent maneuvers produced by the proposed algorithm can be strategically combined to ensure that the obstacles are collectively avoided. We will demonstrate one such implementation in simulations presented in Sections IV-C and V-C.*

IV. Collision Avoidance for Nonholonomic Vehicles

In this section, we consider collision avoidance for vehicles with unicycle kinematics, encompassing a first-order nonholonomic constraint, and input limitations in the form of a maximum turning rate and forward speed. The first part describes the vehicle system, and in the remainder part, we analyze the system with the proposed algorithm in closed-loop, where a set of safety conditions are found and proven to guarantee avoidance. We present simulations of the algorithm applied to a nonholonomic vehicle in two different encounters with multiple dynamic obstacles.

A. Vehicle System

In this section, we present the model and control of the vehicle, including the control objectives and the corresponding nominal guidance law.

1) Model: We consider a nonholonomic vehicle described by

$$\dot{\boldsymbol{p}} = u\left[\cos\psi,\, \sin\psi\right]^{\top}, \tag{18a}$$
$$\dot{\psi} = r, \tag{18b}$$
$$\dot{u} = a, \tag{18c}$$

where $\tau = [r,\, a]^{\top} \in U$ are the turning rate and forward acceleration, representing the control inputs of the system, ψ is the orientation, and $u \in [0, u_{\max}]$ is the forward speed. We collect the state of the vehicle in the state vector $\boldsymbol{x} \triangleq \left[\boldsymbol{p}^{\top},\, \psi,\, u\right]^{\top}$ and define the linear velocity as $\boldsymbol{v} \triangleq \dot{\boldsymbol{p}}$. Moreover, the control inputs are bounded according to

$$U = \{\tau \in \mathbb{R}^2 : |r| \leq r_{\max},\, |a| \leq a_{\max}\}, \tag{19}$$

where the limits $r_{\max}, u_{\max} > 0$ and $a_{\max} \geq 0$ are constant.

2) Control Objectives: The nominal vehicle objective is chosen to be target reaching: The vehicle should reach within a specified circle of acceptance around a target position $\boldsymbol{p}_{\text{t}} \triangleq [x_{\text{t}},\, y_{\text{t}}]^{\top}$. The radius of this circle is given by a parameter $d_{\max} > 0$, representing the maximum allowed distance to the target. Meanwhile, the vehicle should not collide with an obstacle. With $d_{\text{t}} \triangleq \|\boldsymbol{p} - \boldsymbol{p}_{\text{t}}\|$ defining the distance to the target, the objectives can be formulated as the spatial constraints:

$$\begin{aligned} \lim_{t\to\infty} d_{\text{t}} - d_{\max} &\leq 0, \\ \inf_{t\geq 0} d_o - d_{\min} &\geq 0. \end{aligned} \tag{20}$$

3) Nominal Guidance Law: To make the vehicle reach the target position, the nominal direction is given by the pure pursuit guidance law:

$$\psi_{\text{nom}} = \operatorname{atan2}(y_{\text{t}} - y, x_{\text{t}} - x). \tag{21}$$

This direction corresponds to the line-of-sight vector between the vehicle and the target, which yields the shortest way to $\boldsymbol{p}_{\text{t}}$.

4) Controllers: To realize the prescribed objectives, the orientation of the vehicle is controlled, and we apply the kinematic controller:

$$r = \begin{cases} r_{\max} & \text{if } \tilde{\psi} > 0, \\ 0 & \text{if } \tilde{\psi} = 0, \\ -r_{\max} & \text{if } \tilde{\psi} < 0, \end{cases} \tag{22}$$

where $\tilde{\psi} \triangleq \psi_{\text{d}} - \psi$ is the error between the current heading and the desired direction ψ_{d}, and the angular difference is mapped to the interval $\tilde{\psi} \in (-\pi, \pi]$.

Remark 2. *To avoid chattering around $\tilde{\psi} = 0$, the control input* (22) *is in practice implemented as a high-gain proportional controller.*

While the collision avoidance strategy does not require explicit control of the vehicle speed, it is necessary that the speed is within certain bounds:

Assumption IV.1. *If $d_{\text{t}} > d_{\max}$, the vehicle speed is nonzero, $u > 0$, and if $d_o \leq d_{\text{safe}}$, the speed is lower bounded by $u \geq u_{\min}$, where $u_{\min} > u_{o\max}$.*

Remark 3. *An implication of Assumption IV.1 is that the vehicle keeps a higher speed than the obstacle in a near vicinity of it. Since we consider fully non-cooperative obstacles, which may even actively pursue collision, the vehicle must be able to out-maneuver the obstacle in any scenario. This requires it to keep a higher speed, as also noted in Section II. The condition could potentially be relaxed in a more confined environment, but such considerations fall beyond the scope of this study.*

The advantage of decoupling the speed is that the acceleration input can be provided by a separate control loop, e.g. an existing speed controller, which yields a large amount of flexibility in deciding the desired speed trajectory.

B. Safety Conditions

In this part, safety of the proposed strategy is verified through an analysis of the closed-loop system. The first lemma derives a minimum turning requirement which guarantees the vehicle's ability to avoid conflicts before they arise.

Lemma IV.1. *Let Assumption IV.1 hold. Consider a vehicle modeled by* (18)*, with inputs confined to the set $\tau \in U$, and an obstacle modeled by* (12)*. Suppose that $d_o \leq d_{\text{safe}}$. If the vehicle and the obstacle are not currently in a conflict, then the controller* (22) *with $\psi_{\text{d}} = \psi_{\text{ca}}^j$ and j satisfying* (16) *will prevent a conflict if the maximum turning input satisfies*

$$r_{\max} \geq r_{o\max}\frac{u_{o\max}}{u_{\min}} + \frac{a_{o\max}u_{\min} + a_{\max}u_{o\max}}{u_{\min}\sqrt{u_{\min}^2 - u_{o\max}^2}}. \tag{23}$$

Proof. For the nonholonomic vehicle (18), we have $\angle \boldsymbol{v} = \psi$. The angular distances to a conflict are then found in clockwise and counter-clockwise direction, respectively, as [34]

$$\gamma^{\pm} = \pm\psi \mp \psi_{\text{vo}}^{\pm}, \tag{24}$$

where the angles (10) are well-defined under Assumption IV.1. The angular distances are wrapped to the domain $(-2\pi, 2\pi)$ such that they are negative if the vehicle is in a conflict and non-negative otherwise. Thus, the vehicle starts in the (safe) set $S = \{\, \boldsymbol{x} \in D : \gamma^{\pm} \geq 0 \,\}$, with $D = \mathbb{R}^2 \times (-\pi, \pi] \times [0, u_{\max}]$. This set is positively invariant if $\dot{\gamma}^{\pm} \geq 0$ on the boundary $\partial S = \{\, \boldsymbol{x} \in D : \gamma^{\pm} = 0 \,\}$, guaranteeing that the system does not leave this set. According to the control law (22), the avoidance direction (14), and the turning parameter (16), the time-derivative of (24) is for $\gamma^{\pm} = 0$ given by

$$\dot{\gamma}^{\pm} = r_{\max} \mp \dot{\psi}_{\text{vo}}^{\pm}. \tag{25}$$

Let $\theta \triangleq \angle \boldsymbol{r}_{\text{rel}}$ such that $\psi_{\text{cc}}^{\pm} \triangleq \theta \pm \lambda$. The time-derivative of the angle θ is found geometrically as

$$\dot{\theta} = -\frac{\|\boldsymbol{v}_{\text{rel}}\|}{d_o}\sin(\angle \boldsymbol{v}_{\text{rel}} - \theta), \tag{26}$$

and the time-derivative of λ is found by differentiating (3) as

$$\dot{\lambda} = \frac{\|\boldsymbol{v}_{\text{rel}}\|}{d_o}\cos(\angle \boldsymbol{v}_{\text{rel}} - \theta)\tan(\lambda). \tag{27}$$

Since $\gamma^{\pm} = 0 \Leftrightarrow \angle \boldsymbol{v}_{\text{rel}} = \theta \pm \lambda$, we have $\dot{\psi}_{\text{cc}}^{\pm} = 0$. The time-derivative of $\vartheta^{\pm}$ is computed from (11) as

$$\begin{aligned} \dot{\vartheta}^{\pm} = (-r_o + \dot{\psi}_{\text{cc}}^{\pm})\frac{u_o\cos(\varphi^{\pm})}{u\sqrt{u^2 - u_o^2\sin^2(\varphi^{\pm})}} \\ + a_o\frac{\sin(\varphi^{\pm})}{\sqrt{u^2 - u_o^2\sin^2(\varphi^{\pm})}} - a\frac{u_o\sin(\varphi^{\pm})}{u\sqrt{u^2 - u_o^2\sin^2(\varphi^{\pm})}}, \end{aligned} \tag{28}$$

and we use that $\|\boldsymbol{v}\| = u$, $\|\boldsymbol{v}_o\| = u_o$, and $\angle \boldsymbol{v}_o = \psi_o$. Hence,

$$\begin{aligned} \dot{\psi}_{\text{vo}}^{\pm} = -r_o\frac{u_o\cos(\varphi^{\pm})}{u\sqrt{u^2 - u_o^2\sin^2(\varphi^{\pm})}} \\ + a_o\frac{\sin(\varphi^{\pm})}{\sqrt{u^2 - u_o^2\sin^2(\varphi^{\pm})}} - a\frac{u_o\sin(\varphi^{\pm})}{u\sqrt{u^2 - u_o^2\sin^2(\varphi^{\pm})}}, \end{aligned} \tag{29}$$

for $\boldsymbol{x} \in \partial S$. Under Assumption IV.1 and the model (12), an upper bound of $\left|\dot{\psi}_{\text{vo}}^{\pm}\right|$ can be found as the right-hand side of (23), which shows that $\dot{\gamma}^{\pm} \geq 0$ if this condition holds. □

The result of Lemma IV.1 remains valid even when the vehicle accelerates during the evasive maneuver. Furthermore, there exists potential for strategic utilization of the acceleration input to evade collisions, but exploring this aspect is left as a topic for future research. The following theorem presents a summary of the primary contribution made in this section.

Assumption IV.2. *At the initial time, $t = 0$, the distance the vehicle keeps to the obstacle satisfies $d_o(0) \geq d_{\text{safe}}$.*

Theorem IV.1. *Let Assumptions IV.1-IV.2 hold, and consider a vehicle modeled by* (18)*, with inputs confined to the set*

$\tau \in U$, and an obstacle modeled by (12)*. Let condition* (23) *of Lemma IV.1 hold, the safety distance to the obstacle satisfy*

$$d_{\text{safe}} \geq d_{\min} + \frac{2u_{\max} + \pi u_{o\max}}{r_{\max}}, \tag{30}$$

and the maximum distance to the target satisfy

$$d_{\max} \geq \frac{u_{\max}}{r_{\max}}. \tag{31}$$

Then, the vehicle with the control law (22)*, where the reference direction is given by the collision avoidance algorithm* (13)-(17)*, with the nominally desired direction given by the pure pursuit guidance law* (21)*, will remain at least a minimum distance from the obstacle and converge to within a maximum distance of the target, achieving the control objectives* (20)*.*

Proof. It follows from Lemma IV.1 that the vehicle avoids entering a conflict that occurs while $d_o < d_{\text{safe}}$. If there is a conflict as $d_o = d_{\text{safe}}$, then the vehicle will start turning towards the safe direction ψ_{ca}^j at the maximum rate. Since the turning distance cannot be larger than π rad by definition, the time used on the maneuver is upper bounded by $\frac{\pi}{r_{\max}}$, and the distance traveled in the direction of the obstacle is upper bounded by the diameter of the turning circle, $\frac{2u_{\max}}{r_{\max}}$. The distance simultaneously traveled by the obstacle is then at most $\frac{\pi u_{o\max}}{r_{\max}}$. It follows from the safety distance (30) that the vehicle resolves the conflict before a collision can possibly occur. If there is no conflict, it follows from Lemma II.1 that a collision cannot occur. Hence, the vehicle will keep at least a minimum distance to the obstacle.

By Assumption IV.1 the vehicle keeps a higher speed than the obstacle during the avoidance maneuvers. Hence, there exists a time at which the vehicle escapes the obstacle and $\psi_{\text{d}} = \psi_{\text{nom}}$ for all future time. The required turning rate can be found geometrically as $\dot{\psi}_{\text{nom}} = -\frac{u}{d_{\text{t}}}\sin(\psi - \psi_{\text{nom}})$. By condition (31), $r_{\max} > \left|\dot{\psi}_{\text{nom}}\right|$, and hence $\psi = \psi_{\text{nom}}$, for all $d_{\text{t}} > d_{\max}$. The velocity kept by the vehicle is then $\boldsymbol{v} = -u\frac{\tilde{\boldsymbol{p}}}{\|\tilde{\boldsymbol{p}}\|}$, where $\tilde{\boldsymbol{p}} \triangleq \boldsymbol{p} - \boldsymbol{p}_{\text{t}}$. Consider the Lyapunov function candidate $V = \frac{1}{2}\tilde{\boldsymbol{p}}^\top\tilde{\boldsymbol{p}}$. The derivative of V along the trajectories of the system (18) is

$$\dot{V} = \tilde{\boldsymbol{p}}^\top\dot{\tilde{\boldsymbol{p}}} = -u\frac{\tilde{\boldsymbol{p}}^\top\tilde{\boldsymbol{p}}}{\|\tilde{\boldsymbol{p}}\|}, \tag{32}$$

using that $\dot{\boldsymbol{p}} = \boldsymbol{v}$ and $\dot{\boldsymbol{p}}_{\text{t}} = 0$. Assumption IV.1 entails $u > 0$ for all $d_{\text{t}} > d_{\max}$. Hence, $\dot{V}$ is negative definite, and $\tilde{\boldsymbol{p}} \to 0$ while $d_{\text{t}} > d_{\max}$. Consequently, there exists a time at which $d_{\text{t}} = d_{\max}$, concluding the proof. □

C. Simulation Results

This section presents simulations of a nonholonomic vehicle navigating through an obstacle-filled environment by employing the proposed strategy. While the algorithm is intended for avoiding a single obstacle, it can readily be adapted to several obstacles with minor adjustments. In the following simulations, the algorithm was extended to handle multiple dynamic obstacles as follows: When the safety distance to an obstacle is reached, the algorithm produces a safe heading angle by the procedure outlined in Section III. If more obstacles are encountered, this heading angle is adjusted in a similar manner until it represents the closest direction that is safe in terms of all obstacles within the respective safety distances.

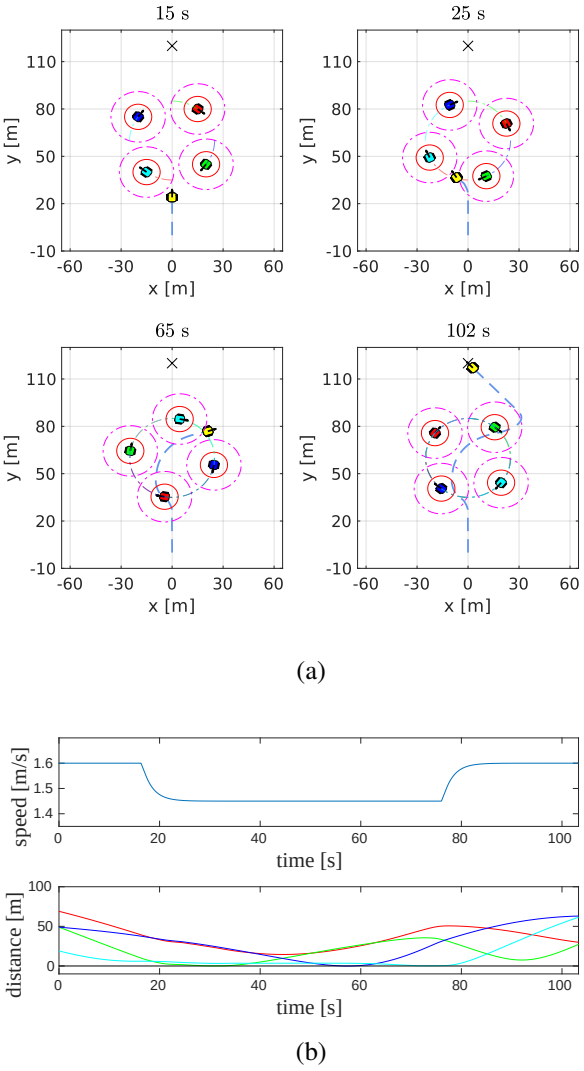


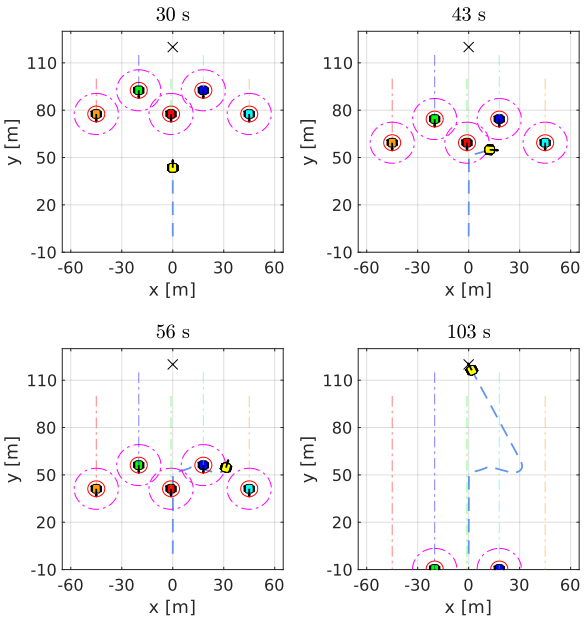
Fig. 5. Snapshots from the first simulation in (a), and the corresponding vehicle speed and distances, $d_o - d_{\min}$, to the obstacles in (b). The vehicle is given by the yellow robot. The remaining robots represent the obstacles, with the red circles marking the obstacles' radii, R_o, and the pink, dashed circles indicating $d_{\min}$. The target is marked by "X".

In both simulations, the vehicle parameters were selected as $d_{\min} = 8 + R_o$ m, with R_o representing the radius of an obstacle, $\psi_{\text{safe}} = \delta = 0.1$ rad, $u_{\min} = 1.45$ m/s, $u_{\max} = 1.6$ m/s, $r_{\max} = 0.8$ rad/s, $a_{\max} = 0.05$ m/s^2, $d_{\max} = 2$ m, and $\boldsymbol{p}_{\text{t}} = [0, 120]^\top$ m. We chose to apply an acceleration of $a_{\text{d}} = -0.5(u - u_{\text{d}})$ s^{-1}, with the desired speed given in the interval $u_{\text{d}} \in [u_{\min}, u_{\max}]$. The obstacle parameters were set to $u_{o\max} = 1.4$ m/s, $a_{o\max} = 0.05$ m/s^2, $r_{o\max} = 0.6$ rad/s. We selected $d_{\text{safe}} = 10 + d_{\min}$ m according to (30).

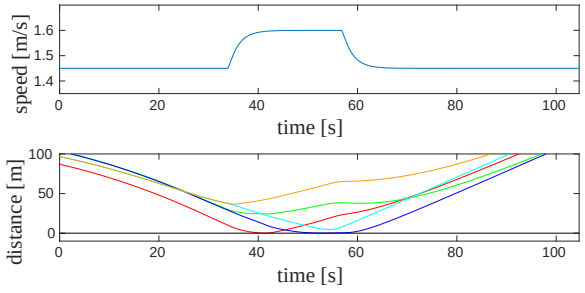
In the first simulation, the vehicle encounters four obstacles of radii $R_o = 8$ m moving in a clockwise circle. Snapshots from the simulation are shown in Figure 5a. The vehicle speed is initialized at $u_{\text{d}} = u_{\max}$, and to illustrate that the vehicle can safely decelerate when applying the algorithm, the desired

speed is lowered to $u_{\mathrm{d}} = u_{\min}$ as the vehicle initiates an evasive maneuver. This can be verified from the vehicle's speed trajectory in Figure 5b. Two consecutive maneuvers are generated by the algorithm, occurring as the vehicle passes between two obstacles at the times 25 s and 65 s, respectively. It can be verified by Figure 5b that the required distances to the obstacles are maintained at all times.

The next simulation features five obstacles of radii $R_o = 5$ m that collectively move parallel to the y-axis, increasing their speeds from $u_o(0) = 0$ towards maximum speed $u_{o\max}$. The configuration of the obstacles is shown in Figure 6a. The vehicle starts at minimum speed, $u_{\mathrm{d}} = u_{\min}$. To demonstrate the strategy also under a positive forward acceleration, we increase the desired speed to $u_{\mathrm{d}} = u_{\max}$ once the safety distance to the closest obstacle is reached, as seen in Figure 6b. The vehicle then turns to the right to avoid two of the obstacles, and it passes between the two outer-most obstacles on the right side before reaching the target. The distances to the obstacles are shown in the bottom plot of Figure 6b, verifying that the required distances to all obstacles are kept.



(a)



(b)

Fig. 6. Snapshots from the second simulation in (a), and the corresponding vehicle speed and distances, $d_o - d_{\min}$, in (b).

V. Collision Avoidance for Underactuated Surface Vessels

In this section, we investigate collision avoidance for marine vessels, specifically for underactuated surface vessels. As opposed to unicycles, such vehicles have a sideways speed component. Due to its underactuation, the vehicle cannot stabilize the sideways motion without losing control authority in the other directions. This section thus presents an extension of the previous theory to a considerably more complex system.

The model of the vehicle is described in the first section, along with the control objectives, the nominal vehicle behavior, and the full control system. An analysis of the proposed collision avoidance algorithm applied to the surface vessel, with a derivation of explicit safety conditions, is given subsequently before we present simulations and experiments underpinning the theoretical results.

A. Vehicle System

In this section, we present the mathematical model of the vessel, the control objectives, and the applied guidance laws and controllers.

1) Model Assumptions:

Assumption V.1. *The vessel is port-starboard symmetric and the body center is located along the center-line of the craft.*

Assumption V.2. *The origin of the body frame is located at the pivot point.*

Remark 4. *Assumption V.1 is satisfied by most marine vehicles by design. Moreover, it is always possible to fulfill Assumption V.2 if the vessel is port-starboard symmetric by performing a coordinate transformation that places the origin of the body frame at the pivot point.*

Assumption V.3. *The hydrodynamic damping is linear.*

Remark 5. *Although nonlinear damping becomes significant at higher speeds, the nonlinear damping terms would only increase the directional stability of the vehicle. It is therefore a common assumption to neglect nonlinear damping to reduce the model complexity.*

2) Vehicle Model: The maneuvering model [35] of a surface vessel can under the assumptions presented in Section V-A1 be expressed as

$$\dot{\boldsymbol{\eta}} = \boldsymbol{R}(\psi)\boldsymbol{\nu}, \tag{33a}$$
$$\dot{u} = f_u(u, v, r) + \tau_u, \tag{33b}$$
$$\dot{v} = X_v(u)r + Y_v(u)v, \tag{33c}$$
$$\dot{r} = f_r(u, v, r) + \tau_r, \tag{33d}$$

where $\boldsymbol{\eta} \triangleq \left[\boldsymbol{p}^\top, \psi\right]^\top$ is the position and orientation of the vehicle with respect to the inertial reference frame, $\boldsymbol{\nu} \triangleq \left[u, v, r\right]^\top$ is the vehicle velocity decomposed in the body-frame, referring to the forward (surge) speed, the sideways (sway) speed, and the yaw (heading) rate, and $\boldsymbol{R} \in \mathrm{SO}(3)$ is the rotation matrix mapping the body-fixed velocities to the inertial frame by

$$\boldsymbol{R}(\psi) \triangleq \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{34}$$

We denote the state of the vehicle as $\boldsymbol{x} \triangleq \left[\boldsymbol{\eta}^\top, \boldsymbol{\nu}^\top\right]^\top$ and the linear velocity as $\boldsymbol{v} \triangleq \dot{\boldsymbol{p}}$. The expressions of the dynamics f_u, f_r, X_v, and Y_v are given in the Appendix and $\boldsymbol{\tau} = [\tau_u, \tau_r]^\top$ represent the control forces in surge and yaw, respectively. The model is valid at maneuvering speeds:

Assumption V.4. *The surge speed of the vehicle lies in the domain*

$$u \in D_u, \quad D_u = [u_{\min}, u_{\max}],$$

where $u_{\max}, u_{\min} > 0$ are constant parameters.

The next conditions ensure that the surface vessel is stable in the lateral direction and maneuverable:

Assumption V.5. *The term Y_v satisfies*

$$Y_v(u) < 0, \quad \forall u \in D_u. \tag{35}$$

Assumption V.6. *The term X_v satisfies*

$$X_v(u) + u > 0, \quad \forall u \in D_u. \tag{36}$$

Remark 6. *Assumption V.5 implies that the vehicle is nominally stable in sway, which can be justified by contradiction: If this is not true, then a small perturbation would lead to the sway speed increasing indefinitely, which is not the case for marine vehicles. Assumption V.6 ensures that a change in the orientation of the vehicle produces a change in the course [30], which is satisfied for most marine vehicles.*

3) Control Objectives: We consider both target reaching and path following to be desired behaviors of the vehicle nominally. In the target reaching case, the objective can be formulated as in Section IV-A2:

$$\lim_{t\to\infty} d_{\mathrm{t}} - d_{\max} \leq 0. \tag{37}$$

The path following objective is for simplicity restricted to straight-line paths on the form $P = \{\, \boldsymbol{p} \in \mathbb{R}^2 : y = y_{\mathrm{Ed}} \,\}$, where y_{Ed} is the desired position along the inertial East axis. The cross-track error to the path is then given by $y_e \triangleq y - y_{\mathrm{Ed}}$, and the path following objective can be expressed as

$$\lim_{t\to\infty} y_e = 0. \tag{38}$$

In both of the above cases, the vehicle should maintain a minimum distance to an obstacle:

$$\inf_{t\geq 0} d_o - d_{\min} \geq 0. \tag{39}$$

4) Nominal Guidance Laws: We employ the pure pursuit guidance law in the target reaching case:

$$\psi_{\mathrm{nom}} = \operatorname{atan2}(y_{\mathrm{t}} - y, x_{\mathrm{t}} - x). \tag{40}$$

For the path following objective, we use a line-of-sight guidance law. The direction corresponds to the line-of-sight vector between the vehicle and a point located a distance $\Delta > 0$ ahead of the vehicle projected onto the path [35]. For the defined path P, the desired direction is thus given by

$$\psi_{\mathrm{nom}} = \operatorname{atan2}(-y_e, \Delta). \tag{41}$$

5) Controllers: The vehicle should keep some positive speed within its maneuvering limits. We let the surge control input be given by

$$\tau_u = -f_u(u, r, v) + a_{\mathrm{d}}, \tag{42}$$

where a_{d} is the applied acceleration. We allow for a_{d} to be given by an arbitrary speed controller under the following condition:

Assumption V.7. *The applied forward acceleration satisfies $|a_{\mathrm{d}}| \leq a_{\max}$, where*

$$a_{\max} < \frac{Y_v^{\min}}{X_v^{\max}} u_{\min}^2, \tag{43}$$

with $Y_v^{\min} = \min_{u\in D_u} |Y_v(u)|$ and $X_v^{\max} = \max_{u\in D_u} |X_v(u)|$.

Moreover, the maintained speed must also remain sufficiently high in the immediate vicinity of the obstacle:

Assumption V.8. *If $d_o \leq d_{\mathrm{safe}}$, the speed of the vehicle is lower bounded by $u \geq U_{\min}$, where*

$$U_{\min} > \max\left\{ u_{o\max}, \sqrt{X_v^{\min}\left(\tfrac{X_v^{\min}}{4} - u_{o\max}\right)} - \tfrac{X_v^{\min}}{2} \right\}, \tag{44}$$

with $X_v^{\min} = \min_{u\in[u_{o\max}, u_{\max}]} X_v(u)$.

Remark 7. *Assumption V.8 serves a similar purpose as Assumption IV.1 for nonholonomic vehicles. Due to the underactuated dynamics, an additional bound on the minimum speed is necessary for marine vessels.*

The remaining control input, τ_r, is chosen to solve the control problems defined in Section V-A3. The sideways motion of the vehicle causes the orientation of the vehicle, ψ, to deviate from the course, denoted by ψ_χ. Since the collision avoidance algorithm outputs direction references, rather than stabilizing the orientation, we will design the yaw controller to stabilize the course, which is possible under Assumption V.6. The course rate, $r_\chi \triangleq \dot{\psi}_\chi$, is found by differentiating the relation $\psi_\chi = \psi + \arctan\left(\frac{v}{u}\right)$ as

$$r_\chi = \frac{(U^2 + X_v(u)u)r + (Y_v(u)u - \dot{u})v}{U^2}, \tag{45}$$

where we substitute the dynamics (33c) and $U \triangleq \sqrt{u^2 + v^2}$ is the vehicle speed. Differentiation of (45) gives

$$\dot{r}_\chi = \frac{\left(\frac{\partial X_v}{\partial u}u + X_v(u)\right)\dot{u}r + \left(\frac{\partial Y_v}{\partial u}u + Y_v(u)\right)\dot{u}v - \ddot{u}v}{U^2} + \frac{(U^2 + X_v(u)u)\dot{r} + (Y_v(u)u - \dot{u})\dot{v}}{U^2} - \frac{2\left(u\dot{u} + v\dot{v}\right)\left(\dot{v}u - \dot{u}v\right)}{U^4}. \tag{46}$$

Substituting the dynamics (33b)-(33d) and the controller (42), the course dynamics can be expressed in feedback-form:

$$\dot{r}_\chi = f_\chi(u, v, r) + g_\chi(u, v)\tau_r, \tag{47}$$

where f_χ and g_χ are given in the Appendix. Since g_χ is nonzero under Assumption V.6 and well-defined under Assumption V.4, we can apply the feedback-stabilizing controller:

$$\tau_r = \frac{1}{g_\chi(u,v)}\left(-f_\chi(u,v,r) + \dot{r}_{\chi\mathrm{d}} - k_r\tilde{r}_\chi\right), \tag{48}$$

where $\tilde{r}_\chi \triangleq r_\chi - r_{\chi\mathrm{d}}$ is the course rate error, $k_r > 0$ is a control gain, and $r_{\chi\mathrm{d}}$ is the desired course rate. Moreover, since we want to stabilize the course angle, we define

$$r_{\chi\mathrm{d}} = \dot{\psi}_\mathrm{d} - k_\chi\tilde{\psi}_\chi, \tag{49}$$

where ψ_d is the desired direction, $\tilde{\psi}_\chi \triangleq \psi_\chi - \psi_\mathrm{d}$ is the course error mapped to the shortest distance $(-\pi,\pi]$, and $k_\chi > 0$ is a control gain. Inserting the control laws (48)-(49) into (47), the course error dynamics can be verified as exponentially stable.

Remark 8. *If the vehicle model suffers from high uncertainty, a more robust method is recommended to control the dynamics of the vehicle, as the above control laws include feedforward terms that may be destabilizing if the model is inaccurate.*

We saturate the desired course rate to ensure that the references assigned to the controller (48) stay bounded when there is a discontinuity in the reference direction:

Assumption V.9. *The desired course rate* (49) *of the marine vessel is saturated such that*

$$|r_{\chi\mathrm{d}}| \leq r_{\max}, \tag{50}$$

where $r_{\max} > 0$ is a constant parameter.

B. Safety Conditions

In this section, we analyze the collision avoidance strategy applied to underactuated marine vessels. An important aspect of this analysis is to address the sway velocity's influence on the vehicle's speed and turning maneuvers. Consequently, our first objective is to establish conditions that guarantee that the sideways speed remains bounded.

Lemma V.1. *Let Assumptions V.1-V.7 hold, and consider a vehicle modeled by* (33)*, with the speed controller* (42)*. Suppose that the course rate of the vehicle is bounded by $|r_\chi(t)| \leq r_{\max}$ for all $t \geq 0$, where $r_{\max}$ is a positive constant. Then, if $|v(0)| \leq v_{\max}$, where*

$$v_{\max} \geq \frac{r_{\max}}{\frac{Y_v^{\min}}{X_v^{\max}} - \frac{a_{\max}}{u_{\min}^2}}, \tag{51}$$

the sway speed is bounded by $|v(t)| \leq v_{\max}$ for all $t \geq 0$.

Proof. Combining (45) with (33c), the course rate can be related to the sway dynamics by

$$\dot{v} = \frac{U^2\left(X_v(u)r_\chi + Y_v(u)v\right) + \dot{u}X_v(u)v}{U^2 + X_v(u)u}. \tag{52}$$

With the speed controller (42), the surge acceleration is given by $\dot{u} = a_\mathrm{d}$, which can be verified by inserting the control law into (33b). Consider the Lyapunov function candidate $V = \frac{1}{2}v^2$. The time-derivative of V along the solutions of (52) is

$$\dot{V} = U^2\frac{X_v(u)r_\chi v + \left(Y_v(u) + X_v(u)\frac{a_\mathrm{d}}{U^2}\right)v^2}{U^2 + X_v(u)u}. \tag{53}$$

Using Assumptions V.4-V.6, we have

$$\dot{V} \leq U^2\frac{|X_v(u)||r_\chi||v| - (|Y_v(u)| - |X_v(u)|\frac{|a_\mathrm{d}|}{u^2})v^2}{U^2 + X_v(u)u}. \tag{54}$$

Assumption V.7 ensures that the quadratic term is negative definite. Moreover, with r_χ bounded, we have

$$\dot{V} \leq \frac{U^2\left(X_v^{\max}r_{\max}|v| - \left(Y_v^{\min} - X_v^{\max}\frac{a_{\max}}{u_{\min}^2}\right)v^2\right)}{U^2 + X_v(u)u}. \tag{55}$$

Hence, for values of v satisfying

$$|v| \geq \frac{r_{\max}}{\frac{Y_v^{\min}}{X_v^{\max}} - \frac{a_{\max}}{u_{\min}^2}}, \tag{56}$$

the quadratic term dominates the linear term, ensuring that $\dot{V} \leq 0$. Consequently, the sublevel set $\Omega_\mathrm{c} = \{v \in \mathbb{R} : V \leq \frac{1}{2}v_{\max}^2\}$, where $v_{\max}$ satisfies (51), is positively invariant. Thus, any trajectory v starting in the set cannot leave it. □

Next, we investigate the impact of the sway speed on the vehicle's required course maneuvers. Through the following lemma, we demonstrate that by ensuring that the maximum course rate, as outlined in Assumption V.9, is set at an adequately high level, the vehicle can effectively and safely avoid any potential conflicts with the obstacle.

Lemma V.2. *Let Assumptions V.1-V.9 hold. Consider a vehicle modeled by* (33) *and an obstacle modeled by* (12)*. Suppose that $d_o \leq d_{\mathrm{safe}}$. If the vehicle and the obstacle are not currently in a conflict, then the controllers* (42) *and* (48)*, with the desired direction given by $\psi_\mathrm{d} = \psi_{\mathrm{ca}}^j$ and j satisfying* (16)*, will prevent a conflict if the vehicle keeps the desired course rate $r_\chi = r_{\chi\mathrm{d}}$ and the maximum course rate satisfies*

$$r_{\max} \geq r_{o\max}\frac{u_{o\max}}{U_{\min}} + \frac{a_{\max}u_{o\max} + U_{\min}a_{o\max}}{U_{\min}\sqrt{U_{\min}^2 - u_{o\max}^2}}. \tag{57}$$

Proof. The first part of the proof follows similar arguments as in the proof of Lemma IV.1. That is, for the marine surface vehicle (33), the direction is given by $\angle\boldsymbol{v} = \psi_\chi$ and the speed by $\|\boldsymbol{v}\| = U$. Hence, the angular distances to a conflict are

$$\gamma^\pm = \pm\psi_\chi \mp \psi_{\mathrm{vo}}^\pm. \tag{58}$$

The distances are non-negative since there is no conflict initially. We want to show that the vehicle remains in the safe set $S = \{\boldsymbol{x} \in D : \gamma^\pm \geq 0\}$, $D = \mathbb{R}^2 \times (-\pi,\pi] \times \mathbb{R}^3$. Consider when the vehicle is on the boundary of this set such that $\gamma^\pm = 0$. Under the control law (49), and with the desired direction given by (14) and (16), the course rate then satisfies

$$r_\chi = \dot{\psi}_{\mathrm{vo}}^\pm \pm k_\chi\delta, \tag{59}$$

where we use that $\dot{\delta} = 0$. By differentiation of (58), we thus have $\dot{\gamma}^\pm \geq 0$ when combined with (59), which renders S positively invariant. It remains to show that this holds when the course rate is bounded. The term $\dot{\psi}_{\mathrm{cc}}^\pm$ vanishes on ∂S, and the time-derivative of $\vartheta^\pm$ (11) is consequently

$$\dot{\vartheta}^\pm = -r_o\varpi_r(\varphi^\pm) + \left(\tfrac{a_o}{u_o} - \tfrac{\dot{U}}{U}\right)\varpi_a(\varphi^\pm), \tag{60}$$

where we define the terms

$$\varpi_r(\varphi) \triangleq \frac{u_o \cos(\varphi)}{\sqrt{U^2 - u_o^2 \sin^2(\varphi)}}, \tag{61a}$$

$$\varpi_a(\varphi) \triangleq \frac{u_o \sin(\varphi)}{\sqrt{U^2 - u_o^2 \sin^2(\varphi)}}, \tag{61b}$$

for conciseness. With $\dot{U} = \frac{u\dot{u}+v\dot{v}}{U}$, the control law (42) and the dynamics (33b), we thus have

$$\dot{\psi}_{\mathrm{vo}}^{\pm} = \frac{a_o}{u_o}\varpi_a(\varphi^{\pm}) - \frac{v\dot{v} + ua_{\mathrm{d}}}{U^2}\varpi_a(\varphi^{\pm}) - r_o\varpi_r(\varphi^{\pm}), \tag{62}$$

by (10). The minimum condition is that $r_\chi = \dot{\psi}_{\mathrm{vo}}^{\pm}$. Combining this with the relations (52) and (62), we find a closed expression for the required course rate as

$$r_\chi = \frac{1}{Q(\varphi^{\pm})}\left((-Y_v(u)v^2 - a_{\mathrm{d}}\left(X_v(u)+u\right))\varpi_a(\varphi^{\pm}) + (U^2 + X_v(u)u)\varpi_o(\varphi^{\pm})\right), \tag{63}$$

where we let $Q(\varphi^{\pm}) \triangleq U^2 + X_v(u)(u + \varpi_a(\varphi^{\pm})v)$ and $\varpi_o(\varphi^{\pm}) \triangleq -r_o\varpi_r(\varphi^{\pm}) + \frac{a_o}{u_o}\varpi_a(\varphi^{\pm})$. The term Q can be bounded from below by

$$Q(\varphi^{\pm}) \geq u^2 + X_v(u)u - |X_v(u)||u_o|, \tag{64}$$

using that $\frac{|v|}{\sqrt{u^2+v^2-u_o^2}} < 1$ under Assumption V.8. Solving (64) for u, it holds that $Q > 0$ for speeds satisfying (44), ensuring that (63) remains well-defined. Since we do not know the contribution of the sway speed v, we analyze the problem using the Lyapunov function candidate $V = \frac{1}{2}v^2$. The time-derivative of V along the trajectories of (52) is

$$\dot{V} = \frac{U^2\left(X_v(u)r_\chi v + Y_v(u)v^2\right) + a_{\mathrm{d}}X_v(u)v^2}{U^2 + X_v(u)u}. \tag{65}$$

Inserting the expression (63) for r_χ yields

$$\dot{V} = \frac{1}{Q(\varphi^{\pm})}\left(v^4Y_v(u) + v^3X_v(u)\varpi_o(\varphi^{\pm}) + v^2(Y_v(u)u^2 + a_{\mathrm{d}}X_v(u)) + vX_v(u)u(a_{\mathrm{d}}\varpi_a(\varphi^{\pm}) + u\varpi_o(\varphi^{\pm}))\right). \tag{66}$$

Notice that Assumption V.7 ensures that the quadratic term is negative definite. Under the model (12) and using Assumption V.8, the term ϖ_a will be bounded by

$$|\varpi_a| \leq \frac{u_{o\max}}{\sqrt{U_{\min}^2 - u_{o\max}^2}}, \tag{67}$$

and similarly the term ϖ_o by

$$|\varpi_o| \leq r_{o\max}\frac{u_{o\max}}{U_{\min}} + \frac{a_{o\max}}{\sqrt{U_{\min}^2 - u_{o\max}^2}}. \tag{68}$$

It follows that for values of v satisfying

$$|v| \geq \frac{\frac{|a_{\mathrm{d}}|}{u}|\varpi_a| + |\varpi_o|}{\frac{|Y_v(u)|}{|X_v(u)|} - \frac{|a_{\mathrm{d}}|}{u^2}} \triangleq v_{\lim} \tag{69}$$

the negative definite terms dominate the terms with indefinite signs, which entails that $\dot{V} \leq 0,\ \forall\, |v| \geq v_{\lim}$. By implication, the course rate must then satisfy

$$|r_\chi| \leq \left(\frac{|Y_v(u)|}{|X_v(u)|} - \frac{|a_{\mathrm{d}}|}{u^2}\right)v_{\lim} = \frac{|a_{\mathrm{d}}|}{u}|\varpi_a| + |\varpi_o|, \tag{70}$$

which follows from the relation (65). Maximizing the above condition then yields the bound (57). □

It is interesting to notice the similarities between the conditions outlined in (23) and (57). Despite the complex underactuated dynamics of surface vessels, the study reveals that the minimum course rate needed to guarantee safety is almost the same as the requirement observed in simpler, unicycle-type vehicles. This insight signifies a significant advancement from the conservative bounds previously reported in [32].

Finally, we present the main findings of our study, summarized in two theorems. These theorems outline sufficient conditions for achieving safe target reaching and path following for underactuated marine vessels.

Assumption V.10. *At the initial time, $t = 0$, the distance kept to the obstacle satisfies $d_o(0) \geq d_{\mathrm{safe}}$, and the course rate and sway speed of the marine vessel satisfy $|r_\chi(0)| \leq r_{\max}$ and $|v(0)| \leq v_{\max}$, respectively, with*

$$v_{\max} \triangleq \frac{r_{\max}}{\frac{Y_v^{\min}}{X_v^{\max}} - \frac{a_{\max}}{u_{\min}^2}}. \tag{71}$$

Theorem V.1. *Let Assumptions V.1-V.10 hold, and consider a vehicle modeled by* (33) *and an obstacle modeled by* (12)*. Let condition* (57) *of Lemma V.2 hold, the safety distance to the obstacle satisfy*

$$d_{\mathrm{safe}} \geq d_{\min} + d_\psi + d_r + d_{\mathrm{ob}}, \tag{72}$$

the safety angle to a conflict satisfy

$$\psi_{\mathrm{safe}} \geq \arccos\left(\frac{d_r}{d_r + d_{\min}}\right), \tag{73}$$

and the maximum distance to the target satisfy

$$d_{\max} \geq \frac{U_{\max}}{r_{\max}}, \tag{74}$$

where

$$d_\psi \triangleq \frac{U_{\max}}{\min\{r_{\max}, k_\chi\frac{\pi}{2}\}}, \qquad U_{\max} \triangleq \sqrt{u_{\max}^2 + v_{\max}^2}, \tag{75a}$$

$$d_r \triangleq (U_{\max} + u_{o\max})t_r, \qquad t_r \triangleq \frac{1}{k_r}\ln\left(\frac{2r_{\max}}{\epsilon}\right), \tag{75b}$$

$$d_{\mathrm{ob}} \triangleq u_{o\max}t_\psi, \qquad t_\psi \triangleq \frac{1}{k_\psi}\ln\left(\frac{\pi}{\delta}\right), \tag{75c}$$

with $0 < \epsilon \ll 1$ sufficiently small. Then, the vehicle governed by the controllers (42) *and* (48)*, where the reference direction is given by the collision avoidance algorithm* (13)-(17)*, with the nominally desired direction given by the pure pursuit guidance law* (40)*, will remain at least a minimum distance from the obstacle and converge to within a maximum distance of the target position, achieving the control objectives* (37),(39)*.*

Proof. Substituting the control law (48) into (47) results in the course rate tracking dynamics $\dot{\tilde{r}}_\chi = -k_r\tilde{r}_\chi$. Since the tracking is both global and exponential, $|r_\chi| \leq r_{\max}$ for all $t \geq 0$ by Assumptions V.9-V.10. It follows from Lemma V.1 that $|v| \leq v_{\max}$ for all $t \geq 0$, and hence $U \leq U_{\max}$.

Suppose that the vehicle must avoid collision due to a potential conflict occurring while $d_o < d_{\mathrm{safe}}$, at some time

$t_0 \geq 0$. Since the desired direction is changed from the nominal reference to (14), there is a jump in ψ_{d}. Under the control laws (48)-(49), the solutions of the course rate tracking satisfy

$$\ln\left(\tilde{r}_\chi(t)/\tilde{r}_\chi(t_0)\right) = -k_r(t - t_0). \tag{76}$$

The time from which the jump occurs until the course rate has converged to within an error ϵ of the desired course rate is thus upper bounded by the time t_r. It follows that $r_\chi \approx r_{\chi\mathrm{d}}$ at some time $t_1 \in [t_0, t_0 + t_r]$. During the convergence, the relative displacement of the obstacle can at most be d_r, which corresponds to the case in which the vehicle and the obstacle move directly towards each other at maximum speeds. It follows from the safety angle (73) that the vehicle cannot enter the conflict during the time t_r, and by Lemma V.2 and II.1 the vehicle avoids a collision with the obstacle.

Suppose at the time t_0, the distance equals $d_o = d_{\mathrm{safe}}$, and a jump occurs in the reference direction. At $t = t_1$, the course rate has converged. Contrary to the previous case, the direction of the vehicle could still be unsafe (conflicting) at this point. With $r_\chi = r_{\chi\mathrm{d}}$, we obtain the course error dynamics from (49):

$$\dot{\tilde{\psi}}_\chi = -k_\chi \tilde{\psi}_\chi. \tag{77}$$

The course error at the time t_1 is upper bounded by π rad by definition. It follows that the time until the course has converged to within an error δ of the desired direction $\psi_{\mathrm{d}} = \psi^j_{\mathrm{ca}}$ is upper bounded by t_ψ. Hence, there exists some time $t_2 \in [t_0, t_0 + t_r + t_\psi]$ at which the vehicle is no longer in conflict with the obstacle. The distance traveled towards the obstacle during the turn is upper bounded by the linear distance traveled when performing the first $\frac{\pi}{2}$ rad turn. The minimum course rate of the vehicle can then be found from (49) and Assumption V.9 as $r_\chi = \min\{r_{\max}, k_\chi \frac{\pi}{2}\}$. Note that this lower bound can be justified by seeing that, as the vehicle moves closer to the obstacle, the collision cone expands, causing the desired course to move farther away. The distance traveled during the $\frac{\pi}{2}$ turn is thus upper bounded by d_ψ, and the distance simultaneously traveled by the obstacle is upper bounded by d_{ob}. It follows from the safety distance (72) that the vehicle does not collide with the obstacle before resolving the conflict. Lemma V.2 and II.1 entails that the vehicle avoids the obstacle thereafter. We have thus established that the vehicle maintains a minimum distance to the obstacle.

Once the vehicle resumes nominal behavior, which is ensured by Assumption V.8, the desired direction is given by the pure pursuit guidance law (40) for all future time. Since $r_{\max} > \frac{U}{d_{\mathrm{t}}} \geq \left|\dot{\psi}_{\mathrm{nom}}\right|$ for all $d_{\mathrm{t}} > d_{\max}$ by (74), it follows that there exists some time at which $\tilde{\psi}_\chi = 0$, from which point the velocity kept by the vehicle is given by

$$\boldsymbol{v} = -U\frac{\tilde{\boldsymbol{p}}}{\|\tilde{\boldsymbol{p}}\|}, \tag{78}$$

and it follows from the equivalent arguments to those provided in the second part of Theorem IV.1 that $d_{\mathrm{t}} \to d_{\max}$ in finite time. □

The last theorem follows directly.

Theorem V.2. *Let Assumption V.1-V.10 hold, and consider a vehicle modeled by* (33) *and an obstacle modeled by* (12)*. Let condition* (57) *of Lemma V.2 hold, conditions* (72) *and* (73) *of Theorem V.1 hold, and the lookahead distance satisfy*

$$\Delta > \frac{U_{\max}}{r_{\max}}. \tag{79}$$

Then, the vehicle governed by the controllers (42) *and* (48)*, where the reference direction is given by the collision avoidance algorithm* (13)-(17)*, with the nominally desired direction given by the line-of-sight guidance law* (41)*, will remain at least a minimum distance from the obstacle and converge to the desired path, achieving the objectives* (38) *and* (39)*.*

Proof. It follows from the arguments provided in the proof of Theorem V.1 that the vehicle maintains at least a minimum distance to the obstacle. With the nominal direction defined by (41), an upper bound of the required course rate is $\left|\dot{\psi}_{\mathrm{nom}}\right| \leq \frac{U}{\Delta} < r_{\max}$. It follows that there exists some time at which $\tilde{\psi}_\chi = 0$, and the desired course is kept thereafter. The result of [36] then entails that the vehicle converges to and stays on the desired path, i.e. $y_e \to 0$ as $t \to \infty$. □

C. Simulation Results

In this section, we present two simulations of a marine vessel under the collision avoidance strategy and proposed control system. We used the model parameters of a light AUV (LAUV) [37], which was simulated on the surface. We used $u_{\min} = 1.5$ m/s, $u_{\max} = 2$ m/s, $r_{\max} = 0.5$ rad/s, and $a_{\max} = 0.2$ m/s^2, yielding $X_v^{\max} = 0.69$, $Y_v^{\min} = 2.33$, $v_{\max} = 0.15$ m/s, and $U_{\max} = 2.01$ m/s. We chose $k_r = 2$, $k_\chi = 1$, $\epsilon = 0.1$, and finally $\delta = 0.17$ rad. The obstacle radius was chosen as $R_o = 10$ m, and the remaining parameters were set to $r_{o\max} = 0.4$ rad/s, $a_{o\max} = 0.1$ m/s^2, and $u_{o\max} = 1.2$ m/s. Moreover, we used $d_{\min} = 15$ m, $d_{\mathrm{safe}} = 27$ m, and $\psi_{\mathrm{safe}} = 0.64$ rad satisfying (72)-(73).

In the first simulation, we demonstrate target reaching as the nominal objective, with $\boldsymbol{p}_{\mathrm{t}} = [0,\ 90]^\top$ m and $d_{\max} = 4.5$ m by (74). Snapshots from the simulation are shown in Figure 7a, and the surge and sway speeds of the vehicle are shown in Figure 7c. We simulated two obstacles, both moving in a constant direction and applying a forward acceleration of $a_o = 0.05$ m/s^2. Note that the vehicle's avoidance maneuvers were combined in the manner described in Section IV-C. Figure 7a shows that the first obstacle approaches the vehicle head-on, and the vehicle makes a left turn to avoid the collision. At this point, an acceleration of $a_{\mathrm{d}} = -(u - u_{\max})$ s^{-1} is applied to demonstrate that safety is upheld also during significant speed changes. The next obstacle approaches from the left side, and the vehicle turns further left to pass behind it. The vehicle maintains the required separation as verified by the bottom plot of Figure 7c and reaches the target position shortly after.

In the second simulation, we considered path following, with $y_{\mathrm{Ed}} = 0$ m and $\Delta = 6$ m. The vehicle was simulated to keep a constant, minimum surge speed and encounter a single obstacle. Figure 7b shows snapshots from the simulation,

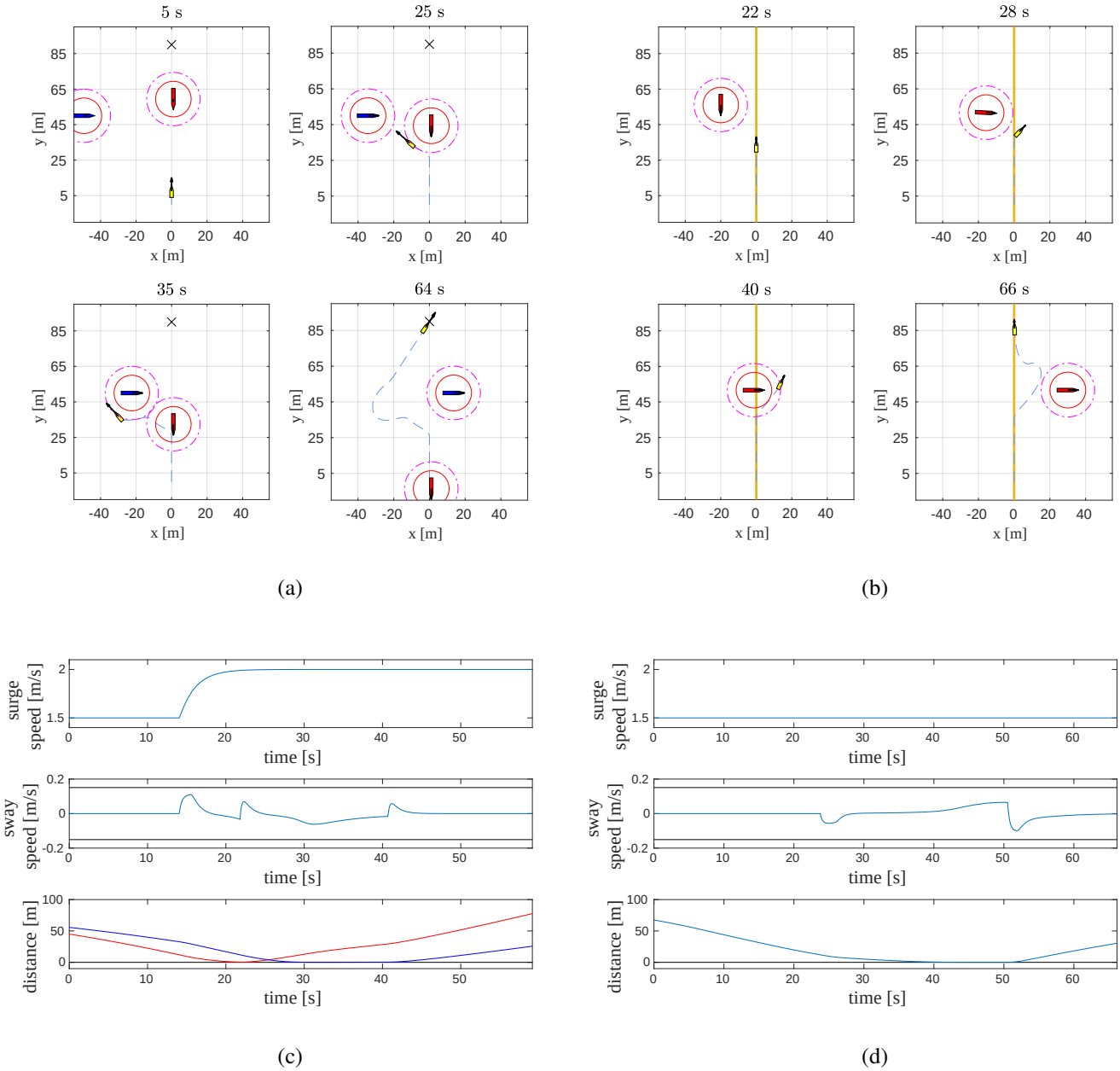


Fig. 7. Snapshots from the first (a) and the second (b) simulation, and the corresponding vehicle surge and sway speeds and distances, $d_o - d_{\min}$, in (c)-(d). The vehicle is the yellow polygon. The remaining polygons represent the obstacles, with R_o and $d_{\min}$ given by the red and pink circles, respectively. The target is marked by "X" and the desired path by the yellow line.

where the obstacle is simulated to first move parallel to the vehicle path and then take a turn in the direction of the vehicle, causing a potential collision. The vehicle leaves the path to evade the obstacle, after which it converges back onto the path. Figure 7d verifies that the vehicle keeps the required distance to the obstacle. Moreover, the vehicle's sway speed stays within the theoretical bound during both simulations.

D. Experimental Results

This section presents the results from two sea trials where the collision avoidance scheme was applied to the LAUV Nemo, shown in Figure 9, in the harbor of Porto, Portugal. The vehicle runs the DUNE (Unified Navigational Environment) [38] onboard software for the control, navigation and logging. It was monitored and commanded using the Neptus interface [38]. The experiments were conducted on the surface, and PID controllers were used for the heading and speed regulation, with direct compensation of the crab angle to ensure tracking of the desired course. The obstacle was simulated on the vehicle onboard software, with a maximum speed of $u_{o\max} = 1.2$ m/s. We used $d_{\min} = 5 + R_o$ m and $d_{\text{safe}} = 16 + d_{\min}$ m, and the commanded vehicle speed was chosen as 1.5 m/s. The remaining vehicle parameters were estimated using a simulation model of the AUV and

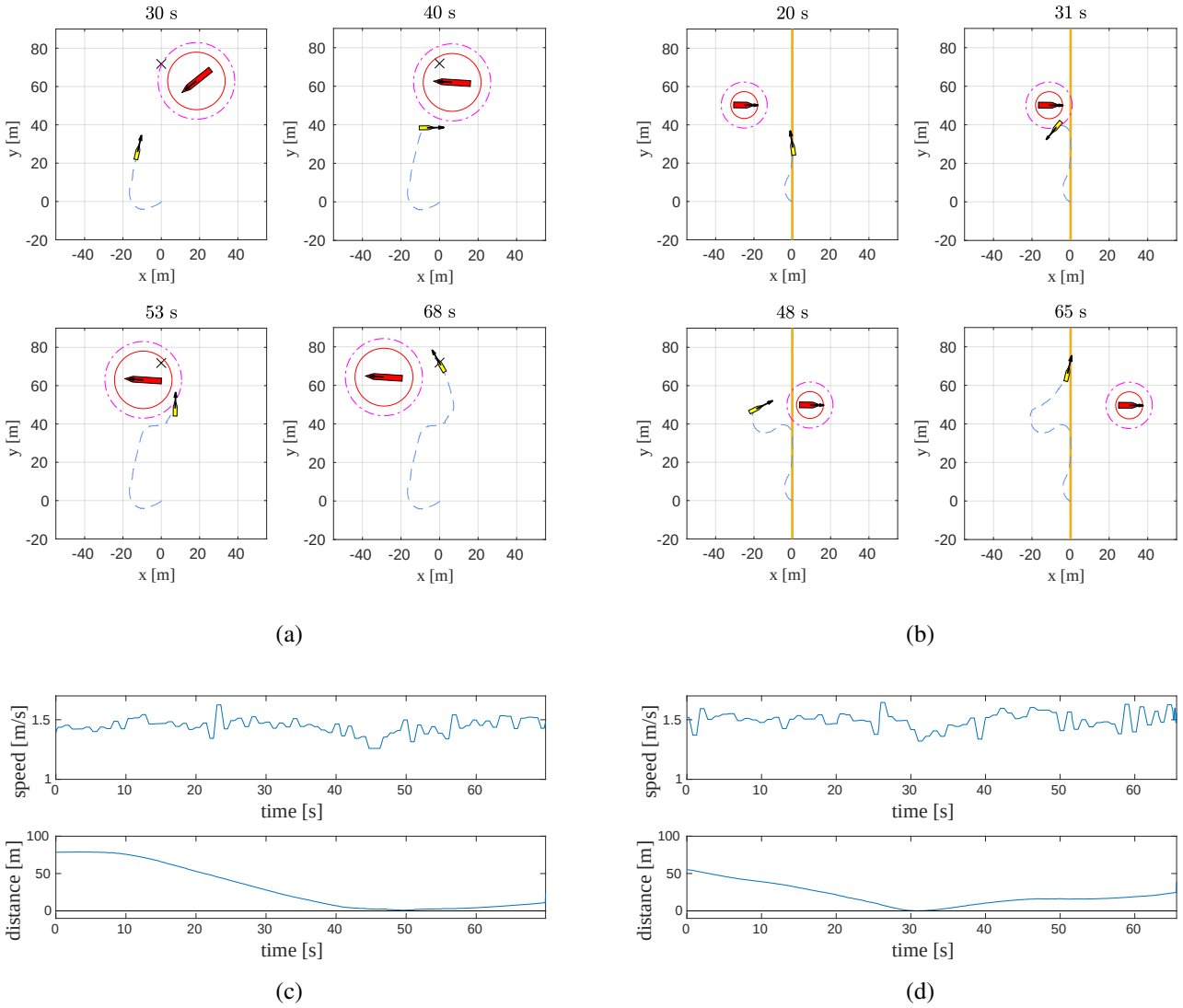


Fig. 8. Snapshots from the first (a) and the second (b) sea trial, and the corresponding estimated speed and distance, $d_o - d_{\min}$, in (c)-(d).

experimental data.

We first applied the collision avoidance strategy in combination with a target reaching nominal goal, where the target position was chosen approximately 75 m from the initial test position. The obstacle was simulated to approach from the right as shown in Figure 8a, with a radius of $R_o = 15$ m. The safety angle was chosen as $\psi_{\text{safe}} = 0.35$ rad. The AUV can be seen to change direction to go behind the virtual obstacle simultaneously as the obstacle takes a turn. The vehicle passes safely and arrives at the target position subsequently. The estimated vehicle speed and the distance to the obstacle during the test are plotted in Figure 8c, which confirms that the vehicle keeps the required distance to the obstacle at all times.

In the second test, the vehicle was to reach the same position but by following a straight-line path to the target. The obstacle was simulated to cross the path from the left, with a smaller radius of $R_o = 8$ m, yielding $d_{\min} = 12$ m and $\psi_{\text{safe}} = 0.45$ rad. The vehicle can be seen to converge to the path in Figure 8b. As the obstacle comes closer, the vehicle must leave the path to avoid a collision. The vehicle evades the obstacle successfully, and it converges back to the path once it no longer leads to a collision. Figure 8d shows that the distance to the obstacle is kept above the required distance during the entire encounter. Thus, in both cases, safety is upheld even with considerable fluctuations in the vehicle's speed.



Fig. 9. The Light Autonomous Underwater Vehicle (LAUV) used to validate the algorithm in a real environment. The torpedo-shaped vehicle measures about 2 meters in length.

VI. Conclusions and Future Work

In this paper, we have presented the analysis, simulations, and experimental validation of a velocity obstacle-based collision avoidance strategy, which requires limited obstacle measurements and is based on simple and intuitive avoidance principles. The strategy is applied to nonholonomic vehicles with input constraints and marine surface vessels with underactuated dynamics, in encounters with moving, non-cooperative obstacles. Through rigorous mathematical studies, we derived explicit conditions on the minimum forward speed and course rate of these vehicles, which together with minimum requirements on the parameters of the algorithm, were shown to guarantee evasion of an obstacle in any potential encounter. The safety criteria were obtained without the requirement of controlling the forward acceleration of the vehicles under some minor conditions on its magnitude, yielding a considerable amount of flexibility in the speed trajectories of the vehicles. We demonstrated how the method adapts to different vehicle tasks by combining it with nominal guidance laws for target reaching and path following, which were proven to be achieved under similar conditions. The theoretical results were supported through simulations of the vehicles in both single and multi-obstacle encounters. To further illustrate the results, the algorithm was experimentally verified through two sea trials with an underactuated marine vessel. The simulation and experimental results demonstrated safety in all test cases, and the nominal control objectives were consistently realized.

Our future work will encompass pursuing an extension of the algorithm to more effectively leverage the vehicle's speed capabilities. While it was shown in simulations that the algorithm can facilitate avoidance of multiple dynamic obstacles, verifying this theoretically and demonstrating it in physical experiments remain other topics for future research.

Acknowledgments

The authors would like to thank Prof. J. Borges de Sousa at the Laboratório de Sistemas e Tecnologia Subaquática (LSTS) for providing access to their underwater vehicle and invaluable support before and during the experiments. We would also like to thank L. Venâncio and P. S. Dias at the LSTS for their guidance on the DUNE onboard software and NEPTUS mission interface. Finally, we wish to thank the operating crew from LSTS for their vital assistance during the sea trials.

Appendix

The functions of the vessel model (33) are defined as

$$
\begin{aligned}
f_u(u,v,r) &\triangleq -\frac{d_{11}}{m_{11}}u + \frac{1}{m_{11}}(m_{22}v + m_{23}r)r,\\
f_r(u,v,r) &\triangleq \frac{m_{23}d_{22} - m_{22}(d_{23} + (m_{22} - m_{11})u)}{m_{22}m_{33} - m_{23}^2}v\\
&\quad + \frac{m_{23}(d_{23} + m_{11}u) - m_{22}(d_{33} + m_{23}u)}{m_{22}m_{33} - m_{23}^2}r,\\
X_v(u) &\triangleq \frac{m_{23}^2 - m_{11}m_{33}}{m_{22}m_{33} - m_{23}^2}u + \frac{d_{33}m_{23} - d_{23}m_{33}}{m_{22}m_{33} - m_{23}^2},\\
Y_v(u) &\triangleq \frac{(m_{22} - m_{11})m_{23}}{m_{22}m_{33} - m_{23}^2}u - \frac{d_{22}m_{33} - d_{32}m_{23}}{m_{22}m_{33} - m_{23}^2},
\end{aligned}
$$

where m_{ij} and d_{ij} denote the elements of the inertia matrix $\boldsymbol{M}$ and the damping matrix $\boldsymbol{D}$, respectively, with

$$
\boldsymbol{M} \triangleq \begin{bmatrix} m_{11} & 0 & 0 \\ 0 & m_{22} & m_{23} \\ 0 & m_{23} & m_{33} \end{bmatrix}, \quad \boldsymbol{D} \triangleq \begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & d_{23} \\ 0 & d_{32} & d_{33} \end{bmatrix}.
$$

The functions of the course dynamics (47) are defined as

$$
\begin{aligned}
f_\chi(u,v,r) &\triangleq \frac{1}{U^2}\left(a_\mathrm{d}\left(\tfrac{\partial X_v}{\partial u}ur + \tfrac{\partial Y_v}{\partial u}uv\right) - v\dot{a}_\mathrm{d} + (X_v(u)ur\right.\\
&\quad \left. + Y_v(u)uv)Y_v(u) + f_r(u,v,r)\left(U^2 + X_v(u)u\right)\right) - \frac{2}{U^4}\left(ua_\mathrm{d}\right.\\
&\quad \left. + v(X_v(u)r + Y_v(u)v)\right)\left((X_v(u)r + Y_v(u)v)u - a_\mathrm{d}v\right),\\
g_\chi(u,v) &\triangleq \frac{U^2 + X_v(u)u}{U^2}.
\end{aligned}
$$

References

[1] M. Hoy, A. S. Matveev, and A. V. Savkin, "Algorithms for collision-free navigation of mobile robots in complex cluttered environments: a survey," *Robotica*, vol. 33, no. 3, p. 463–497, 2015.

[2] J. N. Yasin, S. A. S. Mohamed, M.-H. Haghbayan, J. Heikkonen, H. Tenhunen, and J. Plosila, "Unmanned aerial vehicles (UAVs): Collision avoidance systems and approaches," *IEEE Access*, vol. 8, pp. 105 139–105 155, 2020.

[3] A. Vagale, R. Bye, R. Oucheikh, O. Osen, and T. Fossen, "Path planning and collision avoidance for autonomous surface vehicles II: a comparative study of algorithms," *Journal of Marine Science and Technology*, vol. 26, p. 1307–1323, 2021.

[4] O. Khatib, *The Potential Field Approach And Operational Space Formulation In Robot Control*. Boston, MA, USA: Springer US, 1986, pp. 367–377.

[5] Y. Koren and J. Borenstein, "Potential field methods and their inherent limitations for mobile robot navigation," in *Proc. IEEE Int. Conf. on Robotics and Automation*, Sacramento, CA, USA, 1991, pp. 1398–1404.

[6] D. Panagou, "Motion planning and collision avoidance using navigation vector fields," in *Proc. IEEE Int. Conf. on Robotics and Automation*, Hong Kong, China, 2014, pp. 2513–2518.

[7] A. Singletary, K. Klingebiel, J. Bourne, A. Browning, P. Tokumaru, and A. Ames, "Comparative analysis of control barrier functions and artificial potential fields for obstacle avoidance," in *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems*, Prague, Czech Republic, 2021, pp. 8127–8136.

[8] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in *Proc. European Control Conf.*, Naples, Italy, 2019, pp. 3420–3431.

[9] E. H. Thyri, E. A. Basso, M. Breivik, K. Y. Pettersen, R. Skjetne, and A. M. Lekkas, "Reactive collision avoidance for ASVs based on control barrier functions," in *Proc. IEEE Conf. on Control Technology and Applications*, Montreal, Canada, 2020, pp. 380–387.

[10] M. Keller, C. Haß, A. Seewald, and T. Bertram, "A model predictive approach to emergency maneuvers in critical traffic situations," in *Proc. IEEE Int. Conf. on Intelligent Transportation Systems*, Gran Canaria, Spain, 2015, pp. 369–374.

[11] I. B. Hagen, D. K. M. Kufoalor, E. F. Brekke, and T. A. Johansen, "MPC-based collision avoidance strategy for existing marine vessel guidance systems," in *Proc. IEEE Int. Conf. on Robotics and Automation*, Brisbane, Australia, 2018, pp. 7618–7623.

[12] D. Fox, W. Burgard, and S. Thrun, "The dynamic window approach to collision avoidance," *IEEE Robotics & Automation Magazine*, vol. 4, no. 1, pp. 23–33, 1997.

[13] B.-O. H. Eriksen, M. Breivik, K. Y. Pettersen, and M. S. Wiig, "A modified dynamic window algorithm for horizontal collision avoidance for AUVs," in *Proc. IEEE Conf. on Control Applications*, Buenos Aires, Argentina, 2016, pp. 499–506.

[14] B.-O. H. Eriksen, E. F. Wilthil, A. L. Flåten, E. F. Brekke, and M. Breivik, "Radar-based maritime collision avoidance using dynamic window," in *Proc. IEEE Aerospace Conf.*, Big Sky, MT, USA, 2018, pp. 1–9.

[15] P. Fiorini and Z. Shiller, "Motion planning in dynamic environments using velocity obstacles," *The Int. Journal of Robotics Research*, vol. 17, no. 7, pp. 760–772, 1998.

[16] D. Wilkie, J. van den Berg, and D. Manocha, "Generalized velocity obstacles," in *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems*, St. Louis, MO, USA, 2009, pp. 5573–5578.

[17] J. van den Berg, J. Snape, S. J. Guy, and D. Manocha, "Reciprocal collision avoidance with acceleration-velocity obstacles," in *Proc. IEEE Int. Conf. on Robotics and Automation*, Shanghai, China, 2011, pp. 3475–3482.
[18] J. van den Berg, M. Lin, and D. Manocha, "Reciprocal velocity obstacles for real-time multi-agent navigation," in *Proc. IEEE Int. Conf. on Robotics and Automation*, Pasadena, CA, USA, 2008, pp. 1928–1935.
[19] Y. I. Jenie, E.-J. V. Kampen, C. C. de Visser, J. Ellerbroek, and J. M. Hoekstra, "Three-dimensional velocity obstacle method for UAV deconflicting maneuvers," in *Proc. AIAA Guidance, Navigation, and Control Conf.*, Kissimmee, FL, USA, 2015.
[20] Y. Kuwata, M. Wolf, D. Zarzhitsky, and T. Huntsberger, "Safe maritime autonomous navigation with COLREGS, using velocity obstacles," *IEEE Journal of Oceanic Engineering*, vol. 39, pp. 110–119, 2014.
[21] W. Zhang, S. Wei, Y. Teng, J. Zhang, X. Wang, and Z. Yan, "Dynamic obstacle avoidance for unmanned underwater vehicles based on an improved velocity obstacle method," *Sensors*, vol. 17, no. 12, p. 2742, 2017.
[22] D. K. M. Kufoalor, E. F. Brekke, and T. A. Johansen, "Proactive collision avoidance for ASVs using a dynamic reciprocal velocity obstacles method," in *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems*, Madrid, Spain, 2018, pp. 2402–2409.
[23] Y. Cho, J. Han, J. Kim, P. Lee, and S.-B. Park, "Experimental validation of a velocity obstacle based collision avoidance algorithm for unmanned surface vehicles," in *Proc. IFAC Conf. on Control Applications in Marine Systems, Robotics, and Vehicles*, Daejeon, South Korea, 2019, pp. 329–334.
[24] E. H. Thyri and M. Breivik, "Partly COLREGs-compliant collision avoidance for ASVs using encounter-specific velocity obstacles," in *Proc. IFAC Conf. on Control Applications in Marine Systems, Robotics, and Vehicles*, Kongens Lyngby, Denmark, 2022, pp. 37–43.
[25] G. A. Mercado Velasco, J. E. C. Borst, M. M. van Paassen, and M. Mulder, "The use of intent information in conflict detection and resolution models based on dynamic velocity obstacles," *IEEE Trans. on Intelligent Transportation Systems*, vol. 16, no. 4, pp. 2297–2302, 2015.
[26] J. Alonso-Mora, T. Naegeli, R. Siegwart, and P. Beardsley, "Collision avoidance for aerial vehicles in multi-agent scenarios," *Autonomous Robots*, vol. 39, no. 1, pp. 101–121, 2015.
[27] C. Tan, S. Huang, K. Tan, and R. Teo, "Three dimensional collision avoidance for multi unmanned aerial vehicles using velocity obstacle," *Journal of Intelligent & Robotic Systems*, vol. 97, p. 227–248, 2020.
[28] M. S. Wiig, K. Y. Pettersen, and A. V. Savkin, "A reactive collision avoidance algorithm for nonholonomic vehicles," in *Proc. IEEE Conf. on Control Technology and Applications*, Maui, HI, USA, 2017, pp. 1776–1783.
[29] M. S. Wiig, K. Y. Pettersen, and T. R. Krogstad, "A reactive collision avoidance algorithm for vehicles with underactuated dynamics," in *Proc. IEEE Conf. on Decision and Control*, Melbourne, Australia, 2017, pp. 1452–1459.
[30] M. S. Wiig, K. Y. Pettersen, and T. R. Krogstad, "Collision avoidance for underactuated marine vehicles using the constant avoidance angle algorithm," *IEEE Trans. on Control Systems Technology*, vol. 28, no. 3, pp. 951–966, 2020.
[31] A. Haraldsen, M. S. Wiig, and K. Y. Pettersen, "Vehicle safety of the velocity obstacle algorithm," in *Proc. IEEE Conf. on Decision and Control*, Jeju Island, South Korea, 2020, pp. 5340–5347.
[32] A. Haraldsen, M. S. Wiig, and K. Y. Pettersen, "Reactive collision avoidance for underactuated surface vehicles using the collision cone concept," in *Proc. IEEE Conf. on Control Technology and Applications*, San Diego, CA, USA, 2021, pp. 619–626.
[33] A. Haraldsen, M. S. Wiig, and K. Y. Pettersen, "Dynamic obstacle avoidance for nonholonomic vehicles using collision cones: Theory and experiments," in *Proc. IEEE Conf. on Control Technology and Applications*, Bridgetown, Barbados, 2023, pp. 46–52.
[34] E. Lalish, K. A. Morgansen, and T. Tsukamaki, "Decentralized reactive collision avoidance for multiple unicycle-type vehicles," in *Proc. American Control Conf.*, Seattle, WA, USA, 2008, pp. 5055–5061.
[35] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. John Wiley & Sons, 2011.
[36] T. I. Fossen and K. Y. Pettersen, "On uniform semiglobal exponential stability (USGES) of proportional line-of-sight guidance laws," *Automatica*, vol. 50, no. 11, pp. 2912–2917, 2014.
[37] J. Estrela da Silva, B. Terra, R. Martins, and J. Sousa, "Modeling and simulation of the LAUV autonomous underwater vehicle," in *Proc. IEEE IFAC Int. Conf. on Methods and Models in Automation and Robotics*, Szczecin, Poland, 2007.
[38] J. Pinto, P. S. Dias, R. Martins, J. Fortuna, E. Marques, and J. Sousa, "The LSTS toolchain for networked vehicle systems," in *Proc. MTS/IEEE OCEANS*, Bergen, Norway, 2013, pp. 1–9.



Aurora Haraldsen (GS'20) received the MSc in engineering cybernetics from the Norwegian University of Science and Technology (NTNU), Trondheim, Norway, in 2020. She is currently pursuing a PhD in engineering cybernetics at NTNU, where she works on collision avoidance for underactuated systems, with an emphasis on marine vessels.



Martin Syre Wiig received the MSc and PhD degrees in engineering cybernetics from the Norwegian University of Science and Technology (NTNU), Trondheim, Norway, in 2007 and 2019, respectively. Since 2007, he has been with the Norwegian Defense Research Establishment (FFI), where he is currently the research manager of the Autonomous underwater vehicles (AUV) program. His research background is in autonomy algorithms for uncrewed vehicles, with a particular focus on AUVs and autonomous surface vehicles (ASVs). As the research manager of the AUV program, he focuses on the technological development required for advanced autonomous and semi-autonomous operation of AUVs.



Kristin Ytterstad Pettersen (S'93-M'98-SM'04-F'17) received the MSc and PhD degrees in engineering cybernetics from the Norwegian University of Science and Technology (NTNU), Trondheim, Norway, in 1992 and 1996, respectively. She is a Professor in the Department of Engineering Cybernetics, NTNU, where she has been a faculty member since 1996. She was Head of Department 2011-2013, Vice-Head of Department 2009-2011, and Director of the NTNU ICT Program of Robotics 2010-2013. She is Adjunct Professor at the Norwegian Defence Research Establishment (FFI) and Key Scientist at the CoE for Autonomous Marine Operations and Systems and the VISTA Centre for Autonomous Robotic Operations Subsea. She is a co-founder of the NTNU spin-off company Eelume AS, where she was CEO 2015-2016. She has published four books and more than 300 papers in international journals and conference proceedings. Her research interests focus on nonlinear control of mechanical systems with applications to robotics, with a special emphasis on marine robotics and snake robotics. She was awarded the IEEE Transactions on Control Systems Technology Outstanding Paper Award in 2006 and in 2017. She was awarded an ERC-AdG-2020 Advanced Grant from the European Research Council, and received the IEEE CSS Hendrik W. Bode Lecture Prize in 2020. She is IEEE Control Systems Society Vice President of Member Activities. She is also an elected member of the IEEE Control Systems Society Board of Governors. She holds several board positions in industrial and research companies. She has served as Associate Editor of IEEE Control Systems Magazine, as Associate and Senior Editor of IEEE Transactions on Control Systems Technology as well as Co-Editor-in-Chief for IFAC Mechatronics. She was IEEE CSS Distinguished Lecturer 2019-2022. She is IEEE Fellow, member of the Norwegian Academy of Technological Sciences, and member of the Academy of the Royal Norwegian Society of Sciences and Letters.